



Few-Shot Multiobjective Multitask Optimization

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Outline

- ❑ Research Topic Introduction
 - ❑ Multiobjective Optimization
 - ❑ Few-Shot Challenges
 - ❑ Multitask Optimization
 - ❑ From Multitask Optimization to Parametric Multitask Optimization
- ❑ Problem 1: Finite Few-Shot Single-Objective Multitask Optimization
 - In-Context Guidance: Learning Inter-Task Synergies via Numerical Foundational Models for Few-Shot Multitask Optimization^[1]
- ❑ Problem 2: Finite Few-Shot Multiobjective Multitask Optimization
 - Bayesian Forward-Inverse Transfer for Multiobjective Optimization^[2]
 - Solving Few-Shot Multiobjective Multitask Optimization via Iterative Sequential Transfer^[3]
- ❑ Problem 3: Parametric Single-Objective Multitask Optimization
 - (θ_l, θ_u) -Parametric Multi-Task Optimization: Joint Search in Solution and Infinite Task Spaces^[4]
- ❑ Problem 4: Parametric Multiobjective Multitask Optimization
 - Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling^[5]
- ❑ Conclusions and Potential Future Work

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[2] Tingyang Wei, et al. "Bayesian forward-inverse transfer for multiobjective optimization." International Conference on Parallel Problem Solving from Nature. Cham: Springer Nature Switzerland, 2024.

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[5] Tingyang Wei, Jiao Liu, Abhishek Gupta, Chin Chun Ooi, Puay Siew Tan, Yew-Soon Ong, “Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling”, *IJCAI 2026*.

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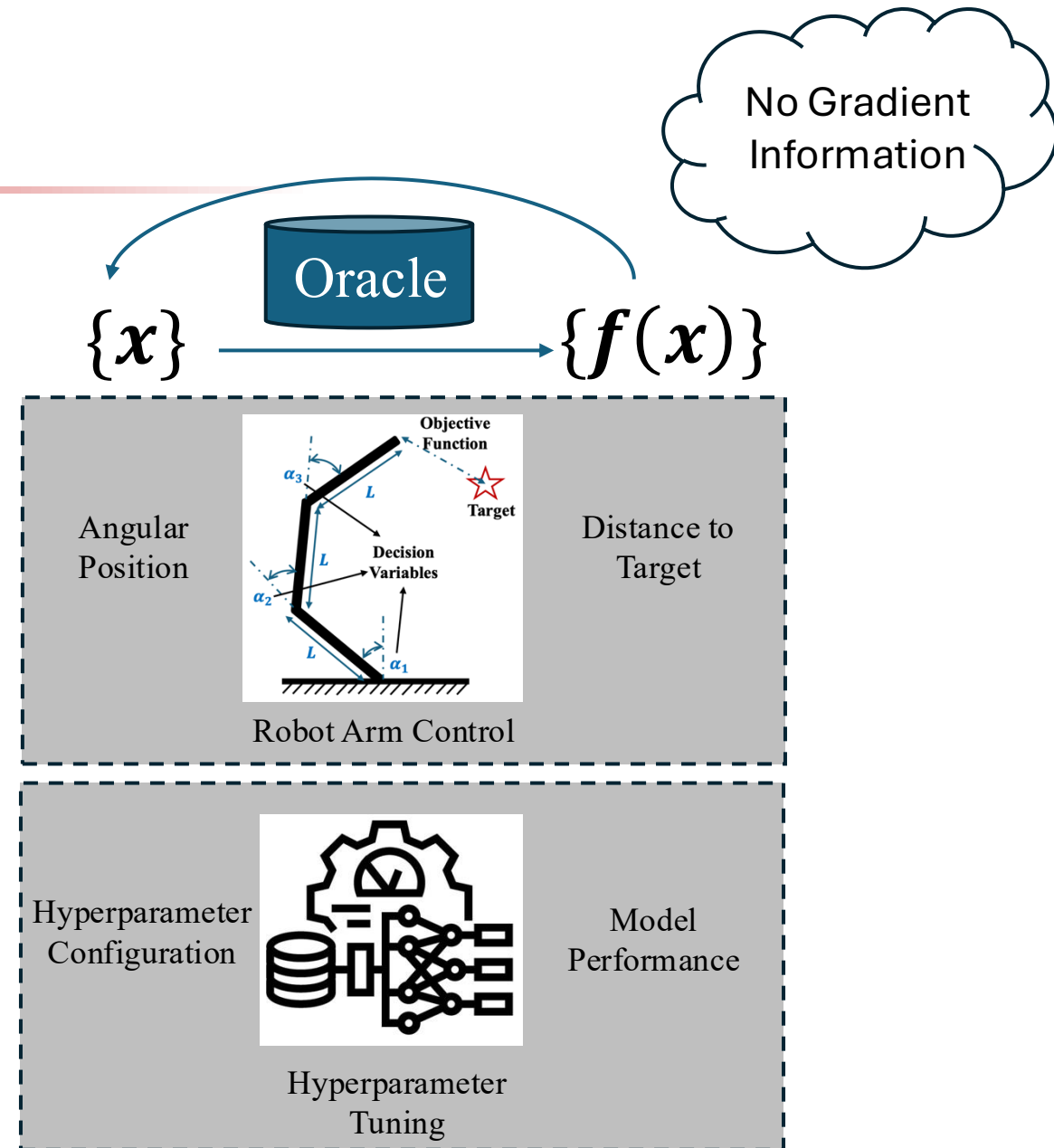
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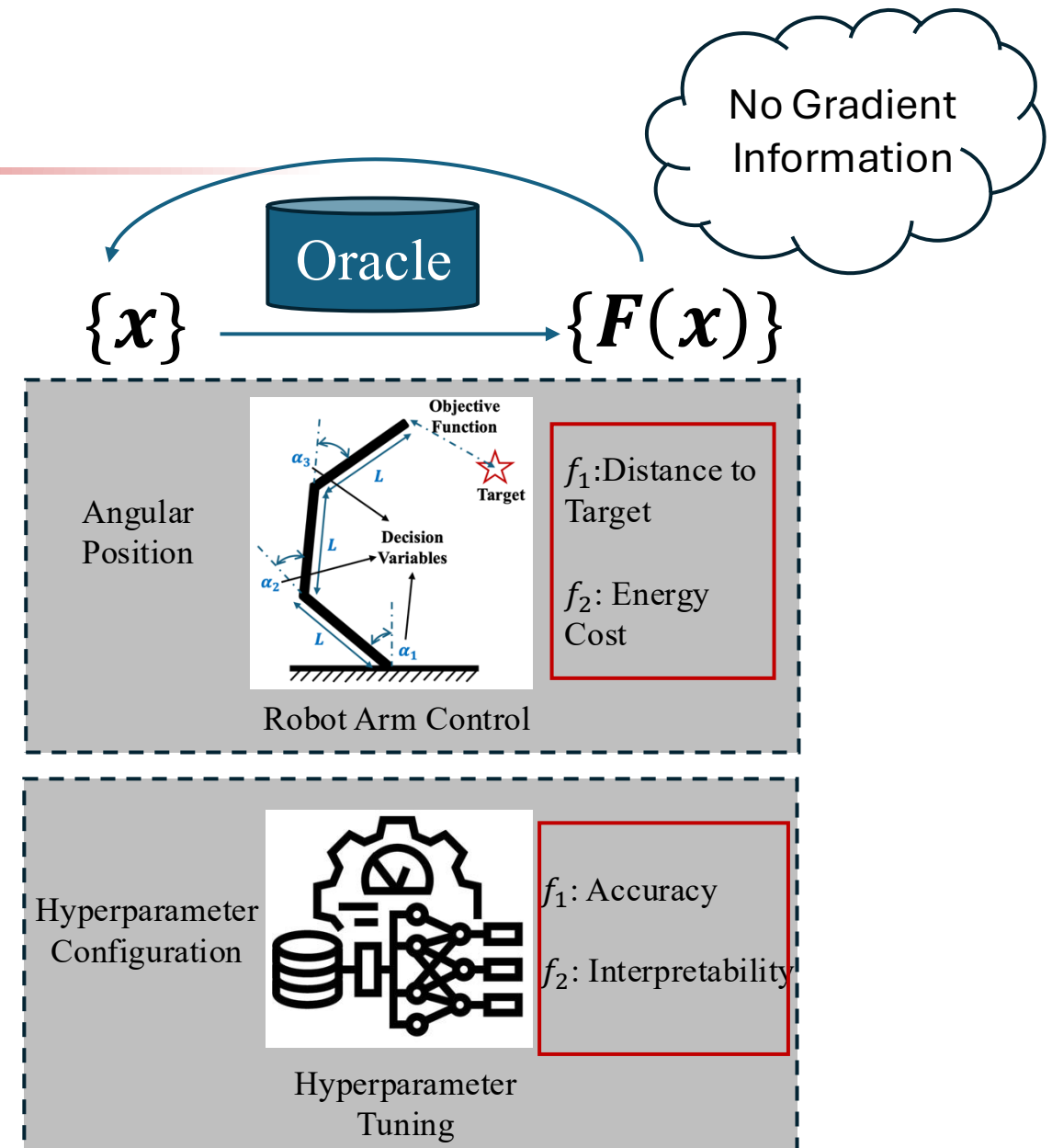
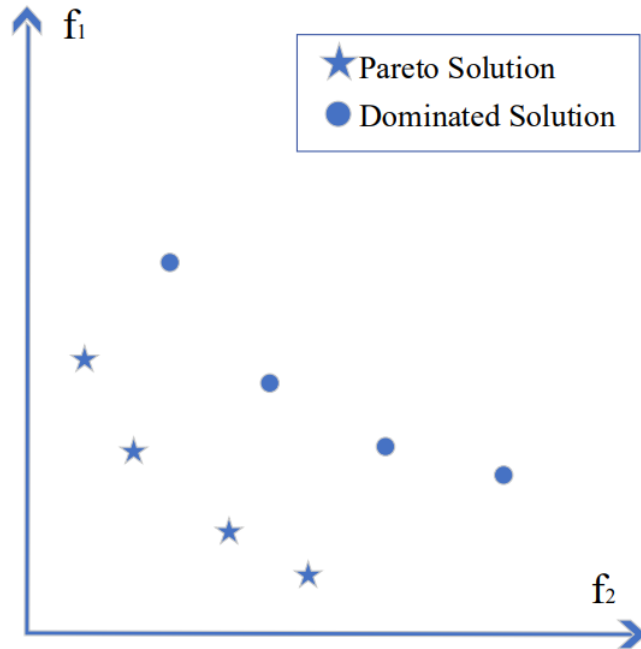
Introduction

- Black-Box Optimization
 - $\operatorname{argmin}_{x \in \mathcal{X}} f(x)$



Introduction

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 - $\operatorname{argmin}_{x \in \mathcal{X}} f(x)$
- Multiobjective Optimization
 - $\operatorname{argmin}_{x \in \mathcal{X}} F(x) = \operatorname{argmin}_{x \in \mathcal{X}} (f_1(x), \dots, f_M(x))$
 - Multiobjective optimization expects **a set of Pareto solutions**,
 - achieving the optimal trade-offs among conflicting objectives.



Introduction

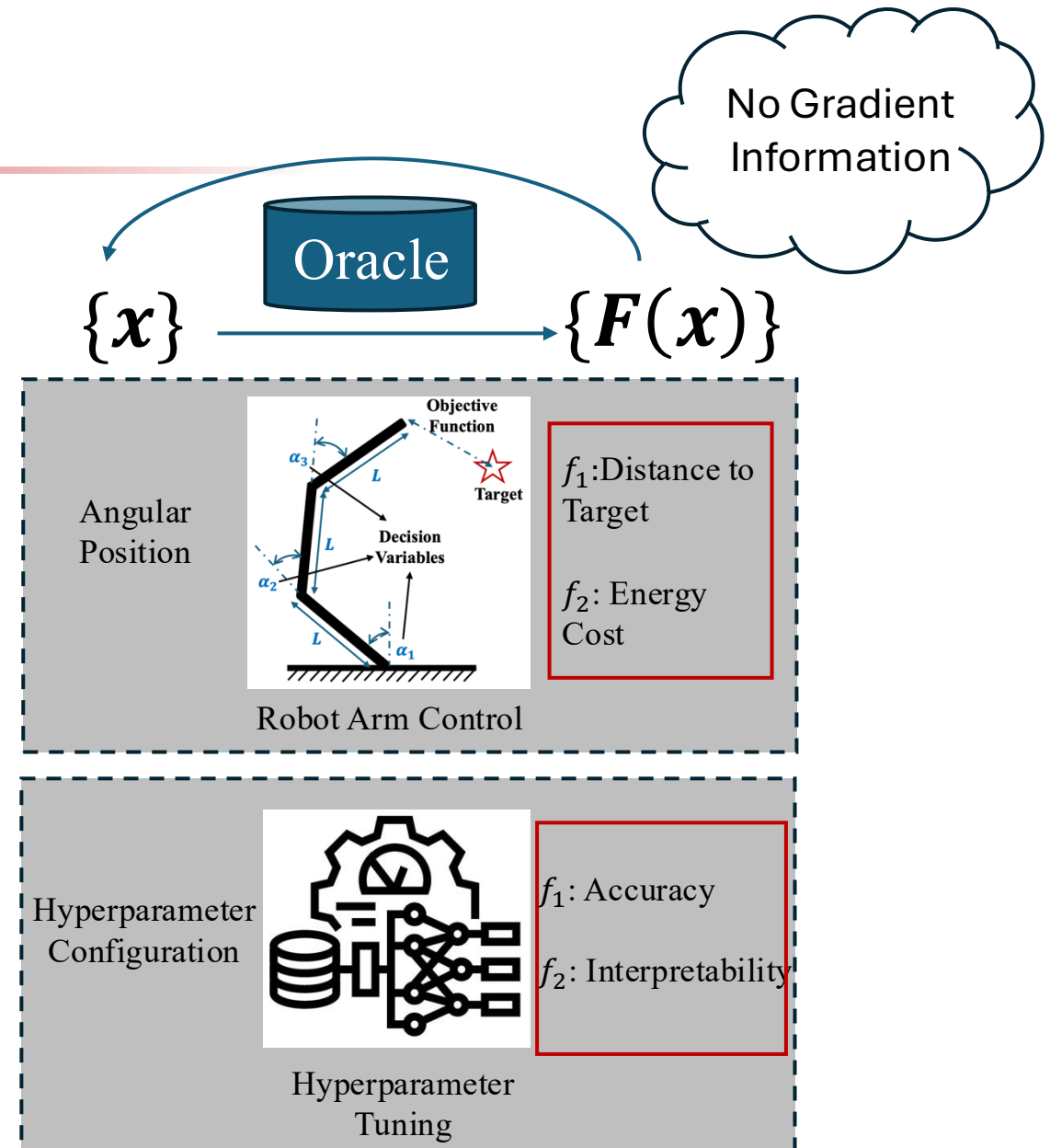
- Black-Box Optimization
 - $\operatorname{argmin}_{x \in \mathcal{X}} f(x)$
- Multiobjective Optimization
 - $\operatorname{argmin}_{x \in \mathcal{X}} F(x) = \operatorname{argmin}_{x \in \mathcal{X}} (f_1(x), \dots, f_M(x))$



- Few-Shot Challenges
 - To conduct (multiobjective) optimization **within limited evaluation budgets**.
 - $\operatorname{argmin}_{x \in \mathcal{X}} f(x)$, $\mathcal{C}_{\max} = \mathcal{C}$
 - $\operatorname{argmin}_{x \in \mathcal{X}} F(x) = \operatorname{argmin}_{x \in \mathcal{X}} (f_1(x), \dots, f_M(x))$, $\mathcal{C}_{\max} = \mathcal{C}$

- Why “Few Shot”?
1. Real problems entail costly function calls.
 2. Problems of other forms (multi-objective) entails even more function calls.

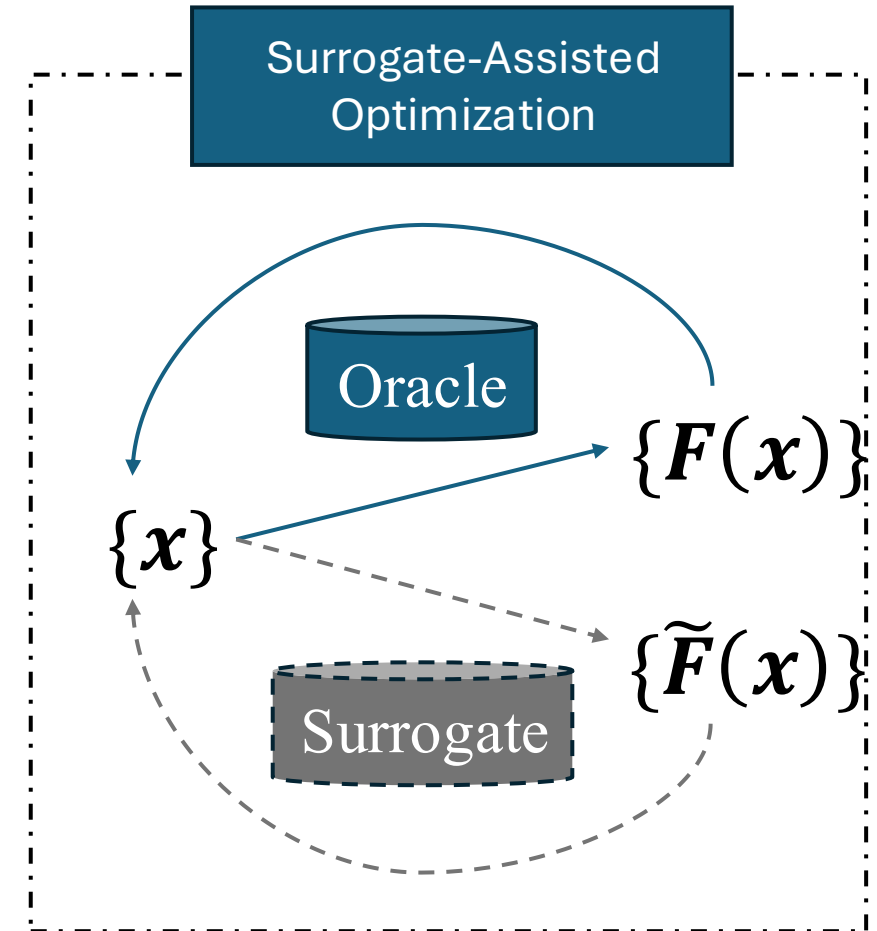
“Few Shot” means budget $\mathcal{C}$ can be really small.



Few-Shot Challenges for Black-Box Optimization

$$\operatorname{argmin}_{x \in \mathcal{X}} f(x), \mathcal{C}_{\max} = \mathcal{C}$$

- With limited function calls, Black-Box Optimization requires building a surrogate model.
 - To approximate the oracle^[1].
 - To inform the next sample to search via uncertainty estimate^[2].
 - In the NeurIPS 2020 Black-Box Optimization Challenge^[3]:
 - “All the top-20 participants used some form of surrogate-assisted optimization.”
 - “This is strongly indicative of the value of using a ‘surrogate-model’”
- During the optimization process, optimizer can query both
 - Surrogates, to gather more information about oracle
 - Oracle, to gather real observations to build better surrogates



[1] Lim, Dudy, et al. "Generalizing surrogate-assisted evolutionary computation." IEEE Transactions on Evolutionary Computation 14.3 (2009): 329-355.

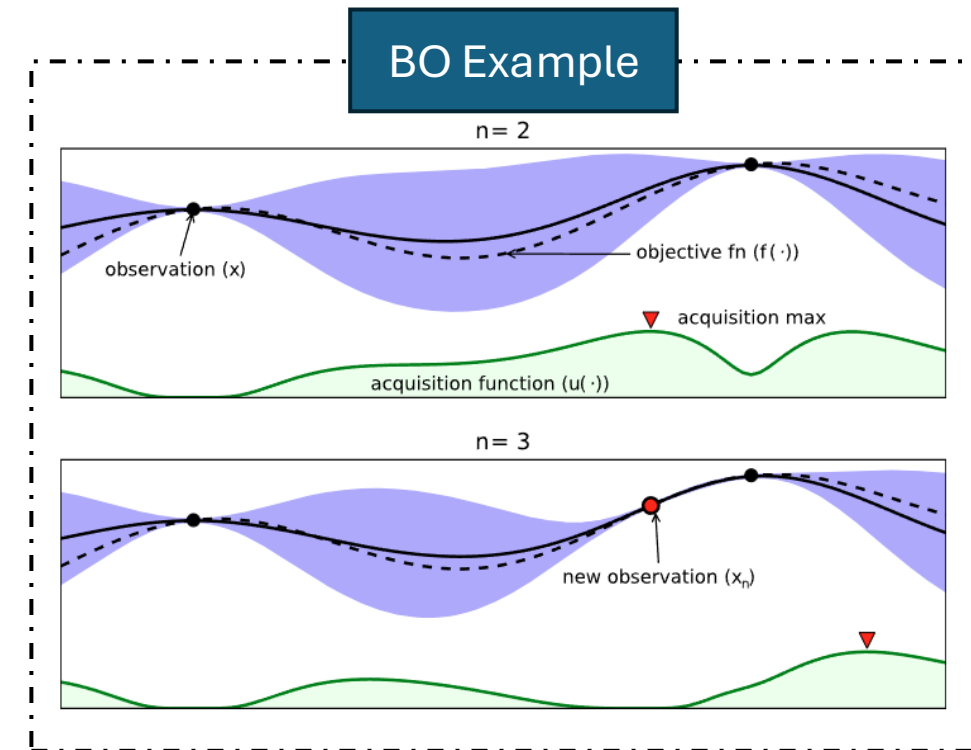
[2] Shahriari, Bobak, et al. "Taking the human out of the loop: A review of Bayesian optimization." Proceedings of the IEEE 104.1 (2015): 148-175.

[3] Turner, Ryan, et al. "Bayesian optimization is superior to random search for machine learning hyperparameter tuning: Analysis of the black-box optimization challenge 2020." NeurIPS 2020 competition and demonstration track. PMLR, 2021.

Few-Shot Challenges for Black-Box Optimization

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- With limited function calls, Black-Box Optimization requires building a surrogate model.
 - In the NeurIPS 2020 Black-Box Optimization Challenge^[1]:
 - “All the top-20 participants used some form of surrogate-assisted optimization.”
 - “This is strongly indicative of the value of using a ‘surrogate-model’... ..”
 - “**All the top teams used some form of Bayesian Optimization**... ..”^[1]
- Bayesian Optimization (BO) is generally considered a default solver in this regard:
 - Iteratively
 - Building Surrogate Model
 - Maximizing the acquisition function to obtain the next sample to search



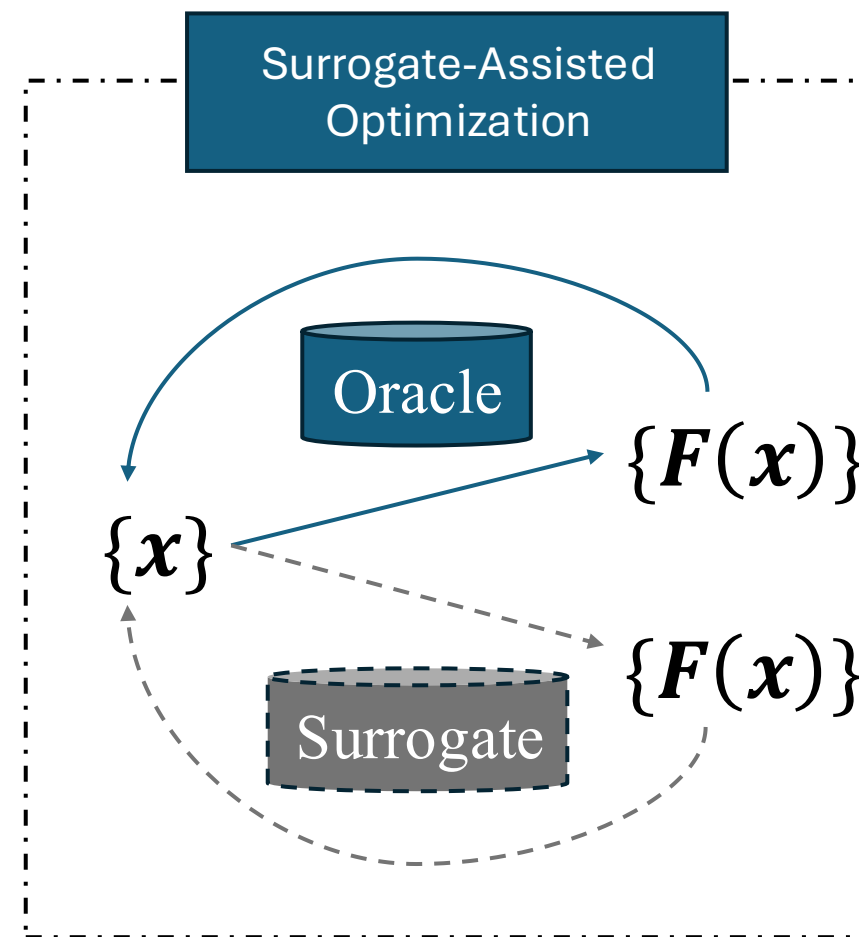
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Few-Shot Challenges for Black-Box Optimization

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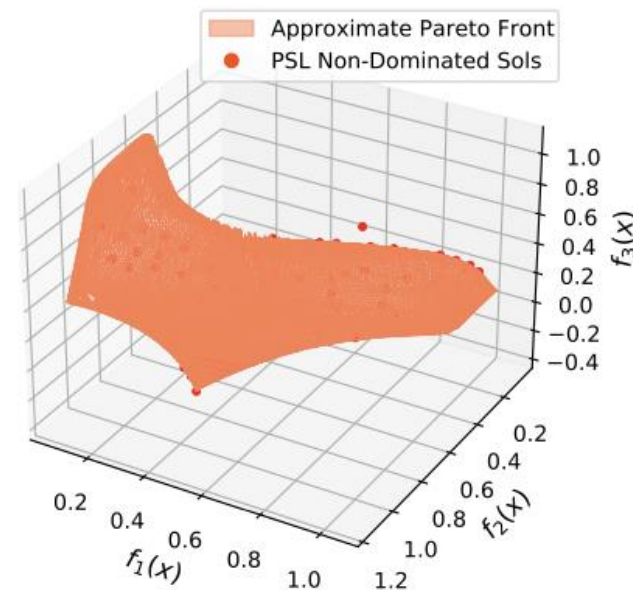
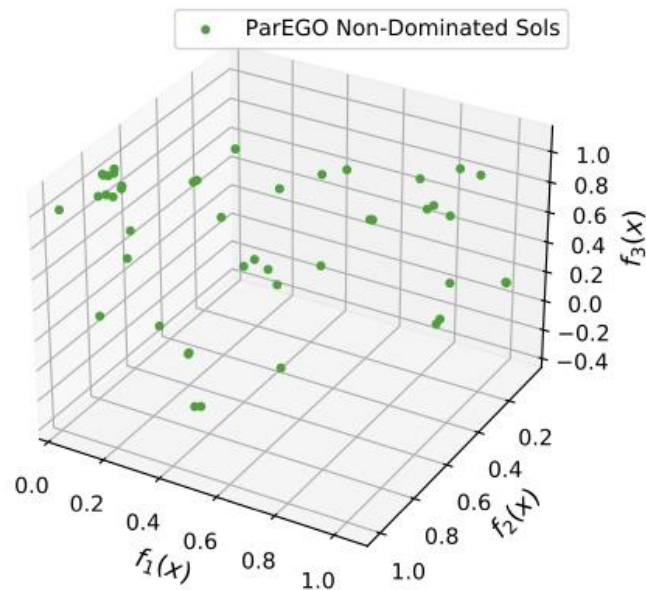
- With limited function calls, Black-Box Optimization requires building a surrogate model.
- Bayesian Optimization (BO) is generally considered a default solver in this regard.
- **However**, the **sparse observations** makes surrogate modeling challenging^[1]:
 - Surrogates can be easily mis-specified.
 - Bias from surrogates can easily skew the potential sampling distributions.



Few-Shot Challenges for Multiobjective Optimization

$$\begin{aligned} \operatorname{argmin}_{x \in \mathcal{X}} F(x) = \\ \operatorname{argmin}_{x \in \mathcal{X}} (f_1(x), \dots, f_M(x)), \\ \mathcal{C}_{\max} = \mathcal{C} \end{aligned}$$

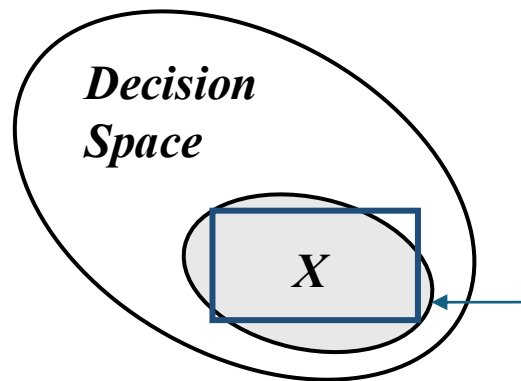
- Few-Shot Multiobjective Optimization expects **a set of Pareto solutions**, achieving the optimal trade-offs among conflicting objectives, **within a limited evaluation budget**.
- Multiobjective optimization (MOO) requires:
 - a large sample size to approximate the **whole Pareto Solution Set**.
 - Few-Shot Settings (a limited number of sample size) make the approximation even more challenging!!!



Few-Shot Challenges for Multiobjective Optimization

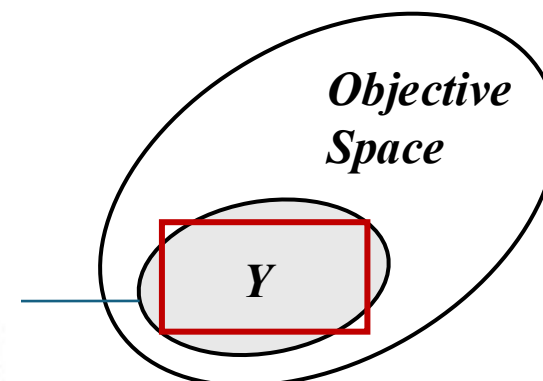
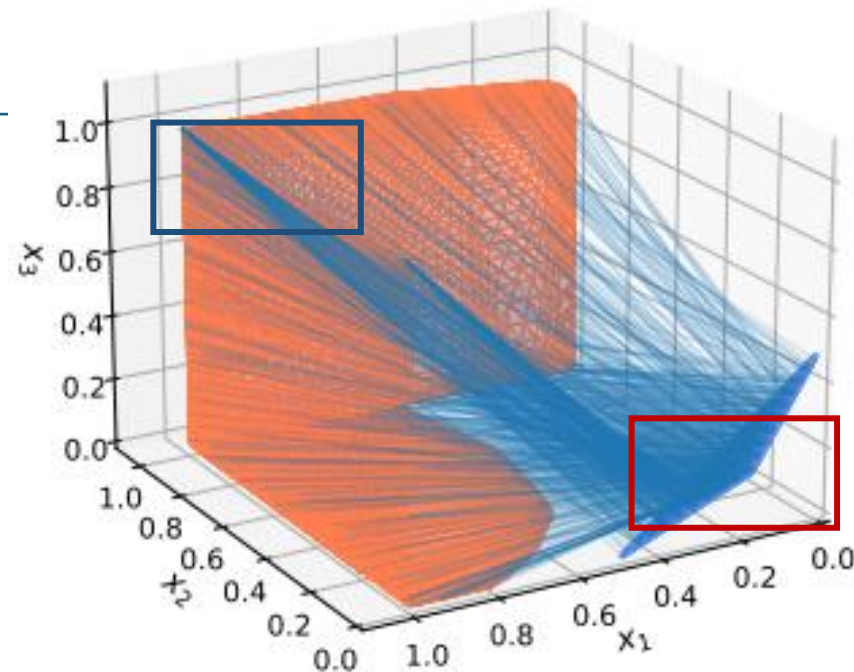
$$\begin{aligned} \operatorname{argmin}_{x \in \mathcal{X}} F(x) = \\ \operatorname{argmin}_{x \in \mathcal{X}} (f_1(x), \dots, f_M(x)), \\ \mathcal{C}_{\max} = \mathcal{C} \end{aligned}$$

- How is the **few-shot** multiobjective optimization resolved?
- Solution:** Inverse Model based multiobjective optimization



During Optimization

X_1
 X_2
 $\dots$
 X_N



Y_1
 Y_2
 $\dots$
 Y_N

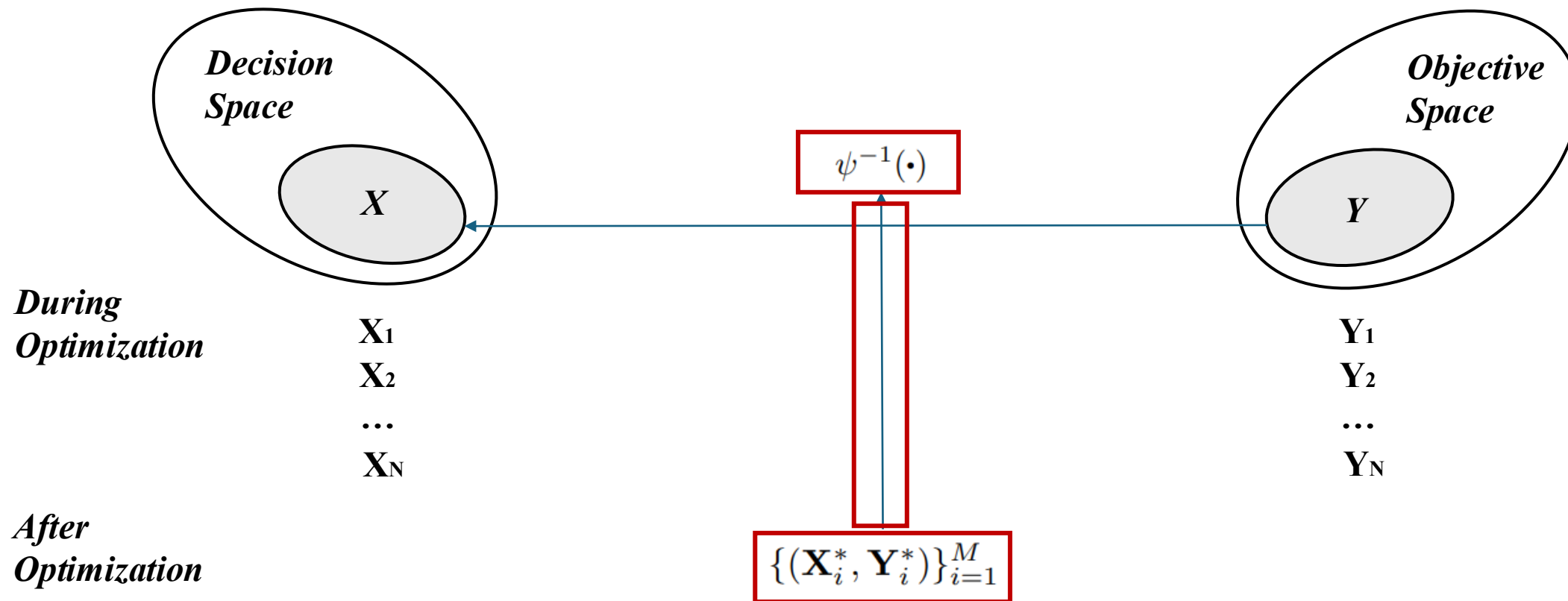
Approximate the Pareto Solution Set with inverse model $\psi^{-1}(\cdot)$

After Optimization

Few-Shot Challenges for Multiobjective Optimization

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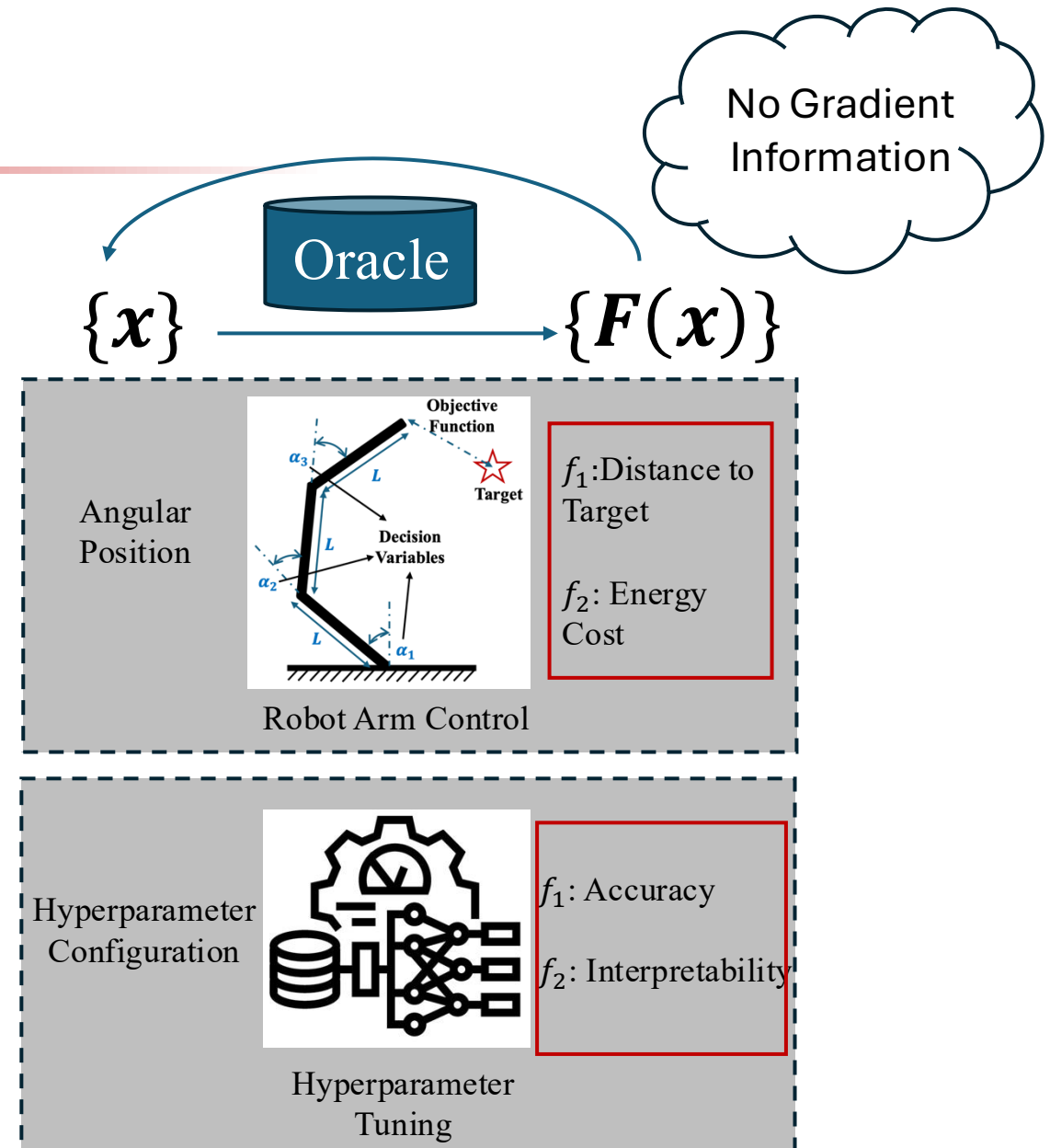
- **Inverse Model** based multiobjective optimization can solve **few-shot** multiobjective optimization.
 - **Limitation 1:** Limited **training data**.
 - **Limitation 2:** The performance of Inverse Model is restricted by the quality of **optimization results**.



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- Few-Shot Challenges
 - To conduct (multiobjective) optimization **within limited evaluation budgets**.
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 - $\operatorname{argmin}_{x \in \mathcal{X}} F(x) = \operatorname{argmin}_{x \in \mathcal{X}} (f_1(x), \dots, f_M(x))$, $\mathcal{C}_{\max} = \mathcal{C}$
- Multitask Optimization
 - $\operatorname{argmin}_{x_m \in \mathcal{X}_m} F_m(x_m)$, $m \in [M]$

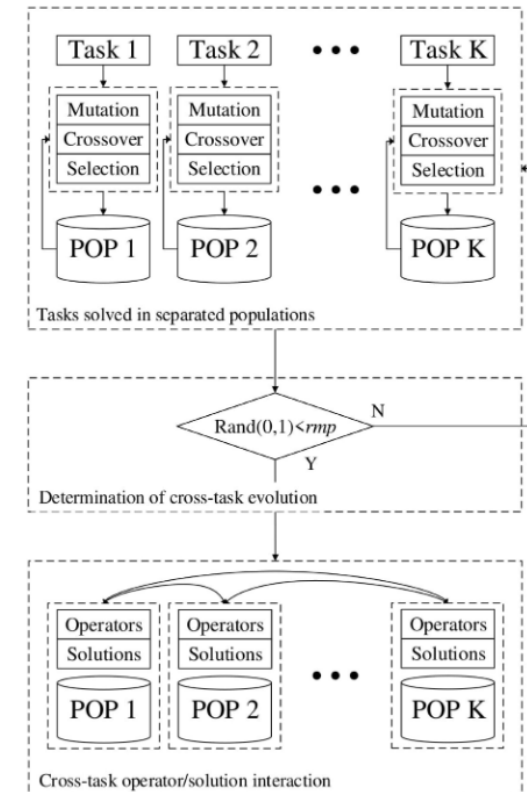
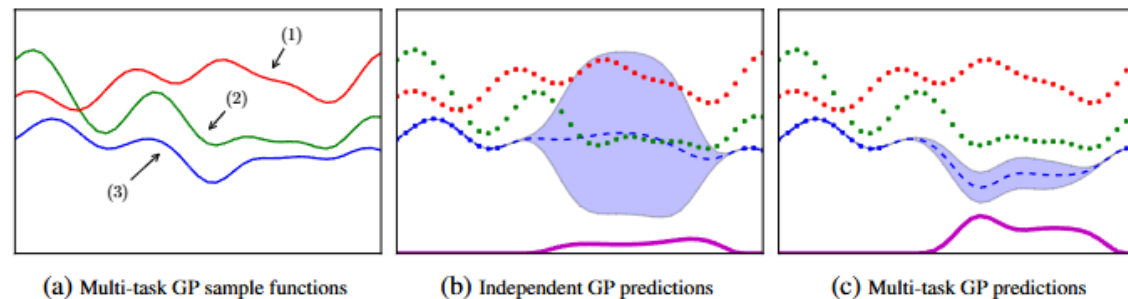
Solve M Optimization
Tasks Simultaneously



Multitask Optimization (How?)

$$\operatorname{argmin}_{x_m \in \mathcal{X}_m} F_m(x_m), m \in [M]$$

- Assuming M problems share certain similarities, we aim to accelerate the convergence of optimization via transferable knowledge of
 - solution distributions,**
 - surrogate models.**
- Solution Distributions (Solution-based Transfer):
 - Evolutionary Multitask Optimization^[1,2]:
 - excavate the power of *implicit parallelism* in EAs and the *knowledge extraction* behavior,
 - through exploiting *the inter-task synergies* across different tasks.
- Surrogate Models (Model-based Transfer):
 - Multitask Bayesian Optimization**^[3]:
 - curate surrogate modeling using *data across different tasks*,
 - analyze *inter-task relationship* via *multitask kernel* in Gaussian Processes.



[1] Gupta, Abhishek, Yew-Soon Ong, and Liang Feng. "Multifactorial evolution: Toward evolutionary multitasking." IEEE Transactions on Evolutionary Computation 20.3 (2015): 343-357.

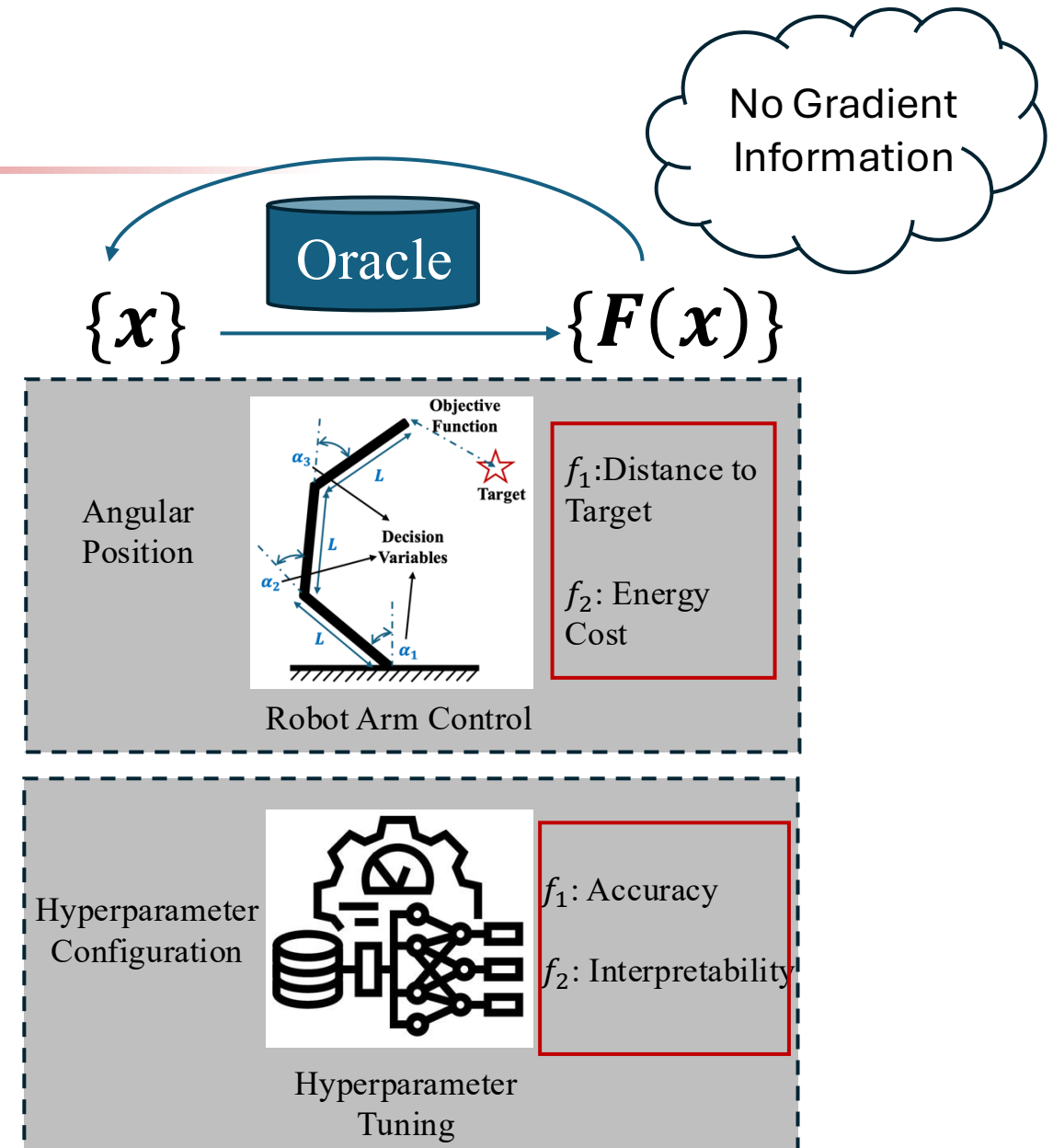
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[3] Swersky, Kevin, Jasper Snoek, and Ryan P. Adams. "Multi-task bayesian optimization." Advances in neural information processing systems 26 (2013).

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- Multitask Optimization (MTO)
 - $\operatorname{argmin}_{x_m \in \mathcal{X}_m} F_m(x_m), m \in [M]$
- Generalizing Multitask Optimization
 - $\operatorname{argmin}_{x \in \mathcal{X}} F(x, \theta), \forall \theta \in \Theta$
 - Multitask Optimization can be considered a special case
 - by instantiating tasks with M task parameters $\{\theta_i\}_{i=1}^M$
 - Parametric Multitask Optimization (PMTO)
 - Single-Objective PMTO
 - Multiobjective PMTO

Each θ characterizes a task, so the space Θ specifies an infinite number of tasks



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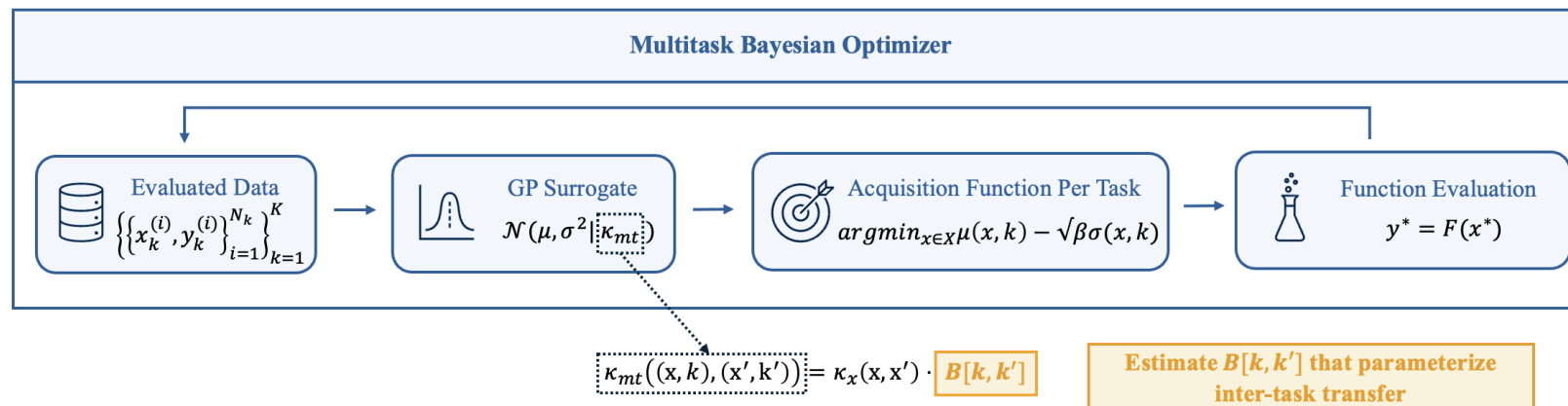
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Few-Shot Multitask Optimization

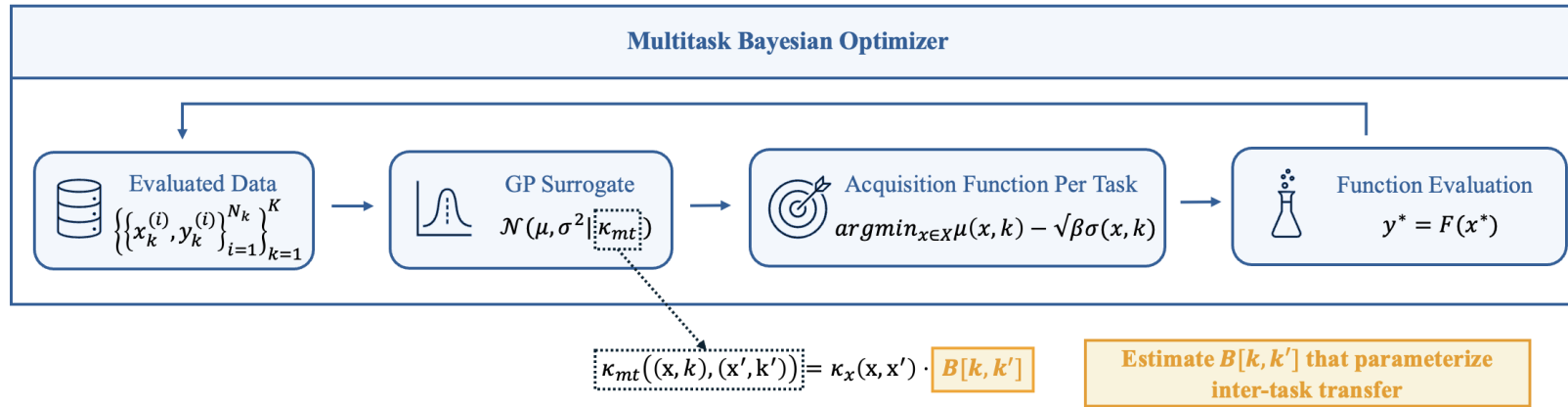
- Few-Shot Optimization:
 - $\operatorname{argmin}_{x \in \mathcal{X}} f(x)$, $\mathcal{C}_{\max} = \mathcal{C}$
 - With limited function calls, **few-shot** optimization requires building a surrogate model to guide the search process.
- Limitations for **few-shot** optimization?
 - Limited **observations** can mis-specify the surrogate model.
- **Multitask** Bayesian Optimizer^[1]:
 - curate surrogate modeling using *data across different tasks*,
 - analyze *inter-task relationship* via *multitask kernel* in Gaussian Processes.



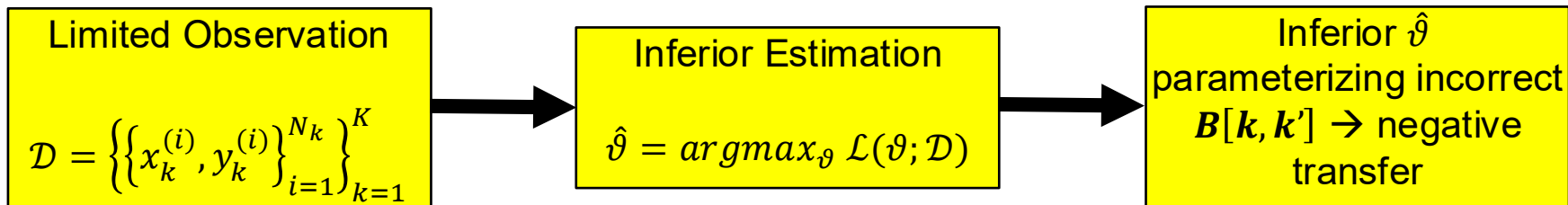
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Few-Shot Multitask Optimization

- Challenges of current Few-Shot Multitask Optimizers:
 - Under few-shot budgets, only a few observations are available per task, making transfer-related components difficult to estimate reliably and increasing the risk of **negative transfer**.



Taking Multitask Bayesian Optimizer for an Example:

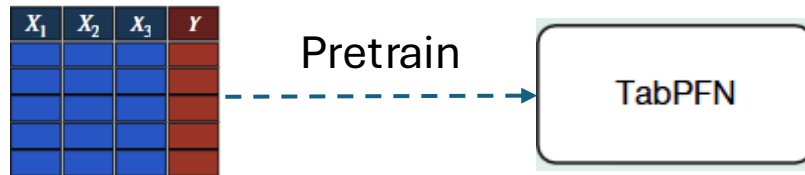


Motivation: TabPFN is a Bayesian Learner in small-data cases

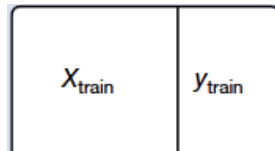
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- TabPFN shows promises in small data classification/regression tasks.

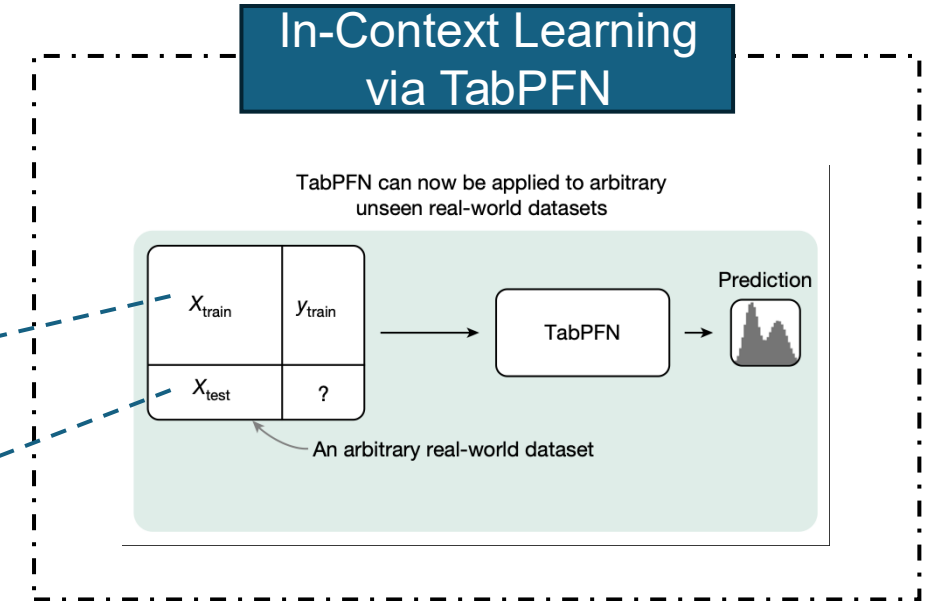
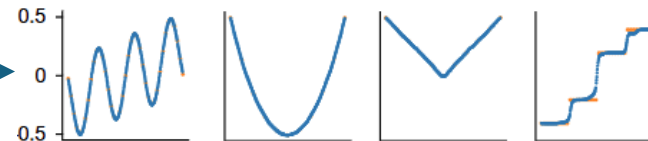
- $P(A)$ (Prior): The pretraining input-output pair
 - Synthetic training data



- $P(B|A)$ (Likelihood): The likelihood of training Data B given pretrained prior A
 - Re-weighting the prior given the training X and Y



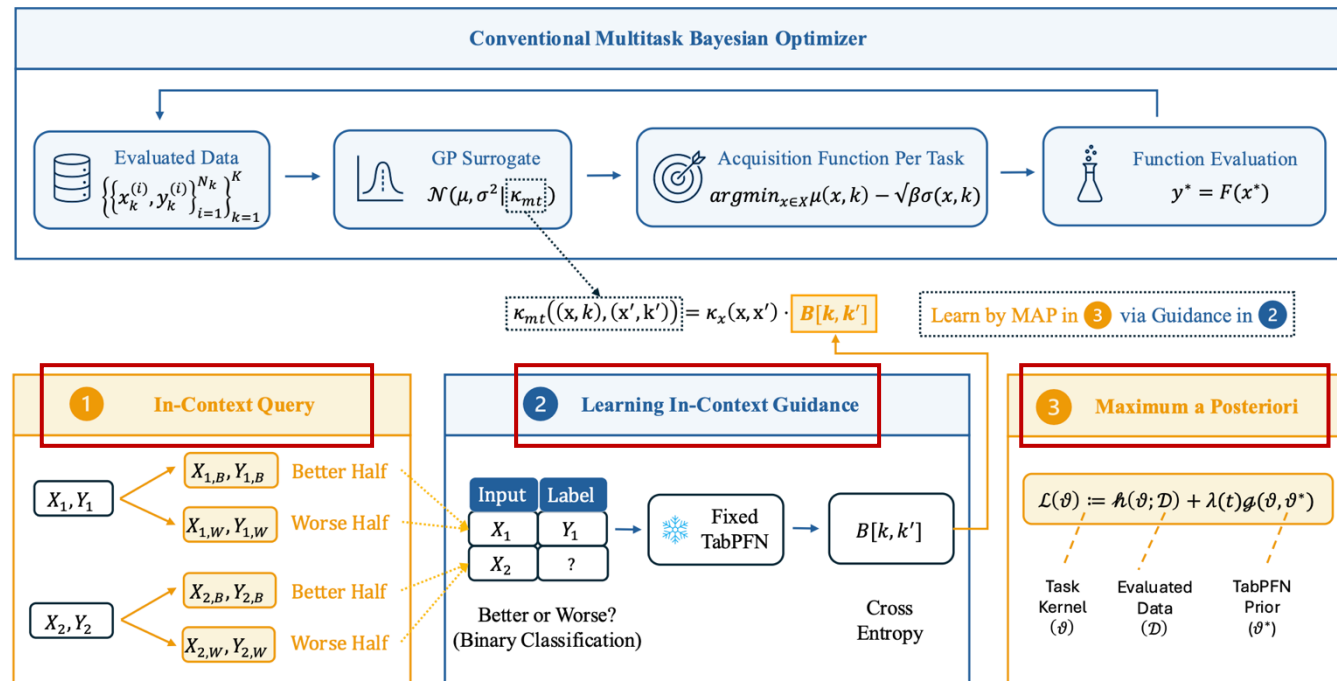
- $P(A|B)$ (Posterior): In-Context Learning Given the context/training-data B.
 - The actual instantiated classifier or regressor given test data



- If so, can TabPFN serve as meta-knowledge to:

- Serve as a **foundational surrogate model** to accelerate optimization process?
- Cater for the optimization in the loop of few-shot multitask optimization?

In-Context Guidance Multitask Optimization: Bayesian Version



In-Context Query

Construct queries for in-context learning.

In MTBO, we query whether TabPFN can specify a better half of solutions of task $\mathcal{T}_2$ given all the solutions of task $\mathcal{T}_1$.

In-Context Guidance Learning

Learn guidance signals via the in-context query.

If TabPFN can specify a better half of solutions of task $\mathcal{T}_2$ given all the solutions of task $\mathcal{T}_1$, the task pair should be similar, vice versa.

MAP Estimation with Foundational Prior

Use guidance signals in MAP estimation.

Treating the in-context guidance as the prior information, the **negative transfer** in small-data estimation can be mitigated for **foundational models**.

In-Context Guidance Multitask Optimization: Bayesian Version

Metric	Evaluations	BO-LCB	MTBO-LCB	BO-LCB-CKT	BO-LCB-BCKT	SELF	ICG-MTBO
+ / - / $\approx$	40	0/16/2	0/14/4	4/12/2	0/17/1	4/6/8	-
	60	0/15/3	1/13/4	3/13/2	0/17/1	1/7/10	-
	80	0/16/2	2/9/7	1/15/2	0/18/0	0/14/4	-
	100	0/15/3	2/8/8	1/15/2	0/18/0	0/14/4	-
Average rank	40	4.94	3.11	3.44	5.11	2.50	1.89
	60	5.06	2.78	3.94	5.00	2.61	1.61
	80	4.83	2.39	4.28	5.06	2.89	1.56
	100	4.78	2.17	4.17	5.06	3.28	1.56

**30-D Problem
with LCB
Acquisition**

Metric	Evaluations	BO-LCB	MTBO-LCB	BO-LCB-CKT	BO-LCB-BCKT	SELF	ICG-MTBO
+ / - / $\approx$	40	0/14/4	0/13/5	0/15/3	0/15/3	0/7/11	-
	60	0/15/3	1/15/2	0/15/3	0/15/3	3/6/9	-
	80	0/15/3	0/14/4	0/15/3	0/17/1	1/10/7	-
	100	0/17/1	0/15/3	0/15/3	0/17/1	1/14/3	-
Average rank	40	4.22	3.00	4.56	4.83	2.72	1.67
	60	4.17	3.11	4.78	5.17	1.94	1.83
	80	4.44	3.00	4.61	5.28	2.17	1.50
	100	4.39	2.56	4.78	5.28	2.61	1.39

**50-D Problem
with LCB
Acquisition**

**Our ICG-MTBO triumphs in
all the settings!!!**

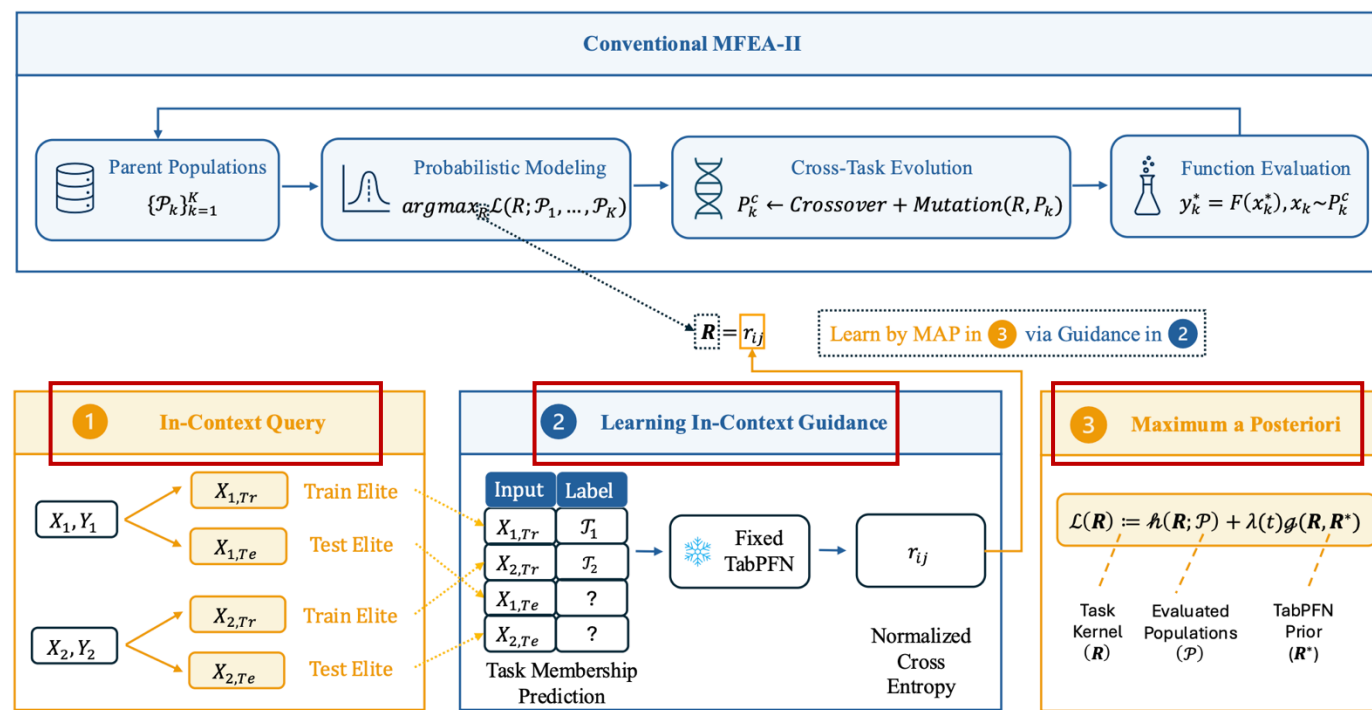
Metric	Evaluations	BO-LogEI	MTBO-LogEI	ICG-MTBO-LogEI
+ / - / $\approx$	40	0/15/3	1/10/7	-
	60	1/14/3	2/11/5	-
	80	1/14/3	3/9/6	-
	100	2/10/6	3/7/8	-
Average rank	40	2.89	1.94	1.17
	60	2.78	2.00	1.22
	80	2.72	1.83	1.44
	100	2.56	1.78	1.67

**30-D Problem
with LogEI
Acquisition**

Metric	Evaluations	BO-LogEI	MTBO-LogEI	ICG-MTBO-LogEI
+ / - / $\approx$	40	2/13/3	1/11/6	-
	60	1/14/3	1/13/4	-
	80	1/14/3	1/14/3	-
	100	0/14/4	2/12/4	-
Average rank	40	2.50	2.11	1.39
	60	2.61	2.11	1.28
	80	2.56	2.17	1.28
	100	2.67	1.94	1.39

**50-D Problem
with LogEI
Acquisition**

In-Context Guidance Multitask Optimization: Evolutionary Version



Remark: ICG is regarded as a general framework as the in-context query can be tailored to the transfer-related parameters for **any multitask optimizer**.

In-Context Query

Construct queries for in-context learning.

In MFEA-II, we query whether TabPFN can specify the task membership among elite solutions of task $\mathcal{T}_2$ and task $\mathcal{T}_1$.

In-Context Guidance Learning

Learn guidance signals via the in-context query.

If TabPFN can distinguish the task membership from gathered elite solutions of task $\mathcal{T}_2$ and task $\mathcal{T}_1$, the task pair should be similar, vice versa.

MAP Estimation with Foundational Prior

Use guidance signals in MAP estimation.

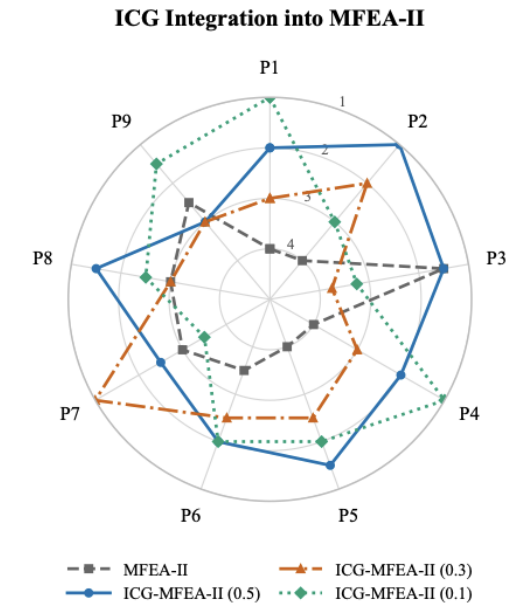
Treating the in-context guidance as the prior information, the **negative transfer** in small-data estimation can be mitigated for **foundational models**.

In-Context Guidance Multitask Optimization: Evolutionary Version

Remark: ICG is regarded as a general framework as the in-context query can be tailored to the transfer-related parameters for **any multitask optimizer**.

We generalize ICG framework to the well-known MFEA-II.

Problem	Task	MFEA-II	ICG-MFEA-II-(0.5)	ICG-MFEA-II-(0.3)	ICG-MFEA-II-(0.1)
P1	1	1.773e+00 (2.2e-01) -	1.395e+00 (5.0e-02) ≈	1.493e+00 (3.1e-01) ≈	1.332e+00 (9.0e-02)
	2	1.168e+03 (3.1e+02) -	8.741e+02 (5.4e+01) -	8.925e+02 (2.9e+02) ≈	7.595e+02 (6.7e+01)
P2	1	1.358e+01 (1.4e+00) -	9.976e+00 (9.2e-01) ≈	1.109e+01 (9.2e-01) ≈	1.115e+01 (6.9e-01)
	2	1.223e+03 (2.4e+02) -	8.047e+02 (7.7e+01) ≈	9.163e+02 (1.6e+02) ≈	9.464e+02 (1.4e+02)
P3	1	2.132e+01 (5.2e-02) ≈	2.131e+01 (4.3e-02) ≈	2.135e+01 (4.6e-02) ≈	2.135e+01 (3.4e-02)
	2	3.968e+03 (3.2e+02) +	4.274e+03 (4.1e+02) ≈	4.495e+03 (4.4e+02) ≈	4.482e+03 (1.6e+02)
P4	1	1.245e+03 (2.3e+02) ≈	9.289e+02 (2.1e+02) ≈	1.016e+03 (1.5e+02) ≈	9.243e+02 (7.2e+01)
	2	2.809e+03 (7.5e+02) ≈	2.018e+03 (7.9e+02) ≈	2.493e+03 (1.2e+03) ≈	1.979e+03 (6.6e+02)
P5	1	1.328e+01 (1.1e+00) -	1.184e+01 (8.4e-01) ≈	1.249e+01 (7.7e-01) -	1.141e+01 (6.7e-01)
	2	2.138e+07 (1.7e+07) -	3.039e+06 (1.8e+06) ≈	5.019e+06 (1.6e+06) ≈	5.355e+06 (3.8e+06)
P6	1	1.308e+01 (1.4e+00) ≈	1.298e+01 (1.1e+00) ≈	1.267e+01 (7.2e-01) ≈	1.298e+01 (1.1e+00)
	2	2.479e+01 (2.9e+00) ≈	2.452e+01 (1.8e+00) ≈	2.499e+01 (2.1e+00) ≈	2.452e+01 (1.8e+00)
P7	1	1.947e+07 (1.0e+07) -	4.415e+06 (3.1e+06) ≈	3.451e+06 (2.5e+06) ≈	4.830e+06 (3.6e+06)
	2	1.087e+03 (1.7e+02) ≈	1.089e+03 (2.0e+02) ≈	1.020e+03 (6.5e+01) ≈	1.123e+03 (1.5e+02)
P8	1	1.751e+00 (1.6e-01) ≈	1.675e+00 (2.2e-01) ≈	1.721e+00 (3.4e-01) ≈	1.673e+00 (2.5e-01)
	2	3.174e+01 (3.1e+00) +	3.170e+01 (4.3e+00) ≈	3.444e+01 (4.8e+00) ≈	3.612e+01 (2.8e+00)
P9	1	1.063e+03 (7.2e+01) ≈	1.330e+03 (1.8e+02) ≈	1.268e+03 (2.0e+02) ≈	1.137e+03 (1.9e+02)
	2	4.597e+03 (7.8e+02) ≈	4.263e+03 (4.7e+02) ≈	4.524e+03 (6.1e+02) ≈	4.158e+03 (7.0e+02)
+/-/≈	-	2/7/9	0/1/17	0/1/17	-
Average rank	-	3.28	1.83	2.61	2.28



Different versions of ICG-MFEA-II can steadily improve MFEA-II.

Outline

- ❑ Research Topic Introduction
 - ❑ Multiobjective Optimization
 - ❑ Few-Shot Challenges
 - ❑ Multitask Optimization
 - ❑ From Multitask Optimization to Parametric Multitask Optimization
- ❑ Problem 1: Finite Few-Shot Single-Objective Multitask Optimization
 - In-Context Guidance: Learning Inter-Task Synergies via Numerical Foundational Models for Few-Shot Multitask Optimization^[1]
- ❑ Problem 2: Finite Few-Shot Multiobjective Multitask Optimization
 - Bayesian Forward-Inverse Transfer for Multiobjective Optimization^[2]
 - Solving Few-Shot Multiobjective Multitask Optimization via Iterative Sequential Transfer^[3]
- ❑ Problem 3: Parametric Single-Objective Multitask Optimization
 - (θ_l, θ_u) -Parametric Multi-Task Optimization: Joint Search in Solution and Infinite Task Spaces^[4]
- ❑ Problem 4: Parametric Multiobjective Multitask Optimization
 - Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling^[5]
- ❑ Conclusions and Potential Future Work

[1] Tingyang Wei, et al. “In-Context Guidance: Learning Inter-Task Synergies via Numerical Foundational Models for Few-Shot Multitask Optimization.” In submission to IEEE Transactions on Evolutionary Computation (2026)

[2] Tingyang Wei, et al. "Bayesian forward-inverse transfer for multiobjective optimization." International Conference on Parallel Problem Solving from Nature. Cham: Springer Nature Switzerland, 2024.

[3] Tingyang Wei, et al. “Solving Few-Shot Multiobjective Multitask Optimization via Iterative Sequential Transfer.” 2026 IEEE Congress on Evolutionary Computation (CEC).

[4] Tingyang Wei, Jiao Liu, Abhishek Gupta, Puay Siew Tan, Yew-Soon Ong, “ (θ_l, θ_u) -Parametric Multi-Task Optimization: Joint Search in Solution and Infinite Task Spaces” IEEE Transactions on Evolutionary Computation (2025).

[5] Tingyang Wei, Jiao Liu, Abhishek Gupta, Chin Chun Ooi, Puay Siew Tan, Yew-Soon Ong, “Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling”, *IJCAI* 2026.

Why Few-Shot Multiobjective Multitask Optimization?

- Few-Shot Multiobjective Optimization:

- $\operatorname{argmin}_{x \in \mathcal{X}} F(x) = \operatorname{argmin}_{x \in \mathcal{X}} (f_1(x), \dots, f_M(x)), \mathcal{C}_{\max} = \mathcal{C}$

$$\{(\mathbf{X}_i^*, \mathbf{Y}_i^*)\}_{i=1}^M \longrightarrow \psi^{-1}(\cdot)$$

- Limitations for **few-shot** multiobjective optimization?

- **Limitation 1:** Limited **training data**.
 - **Limitation 2:** Inverse Model's performance restricted by the quality of **optimization results**.

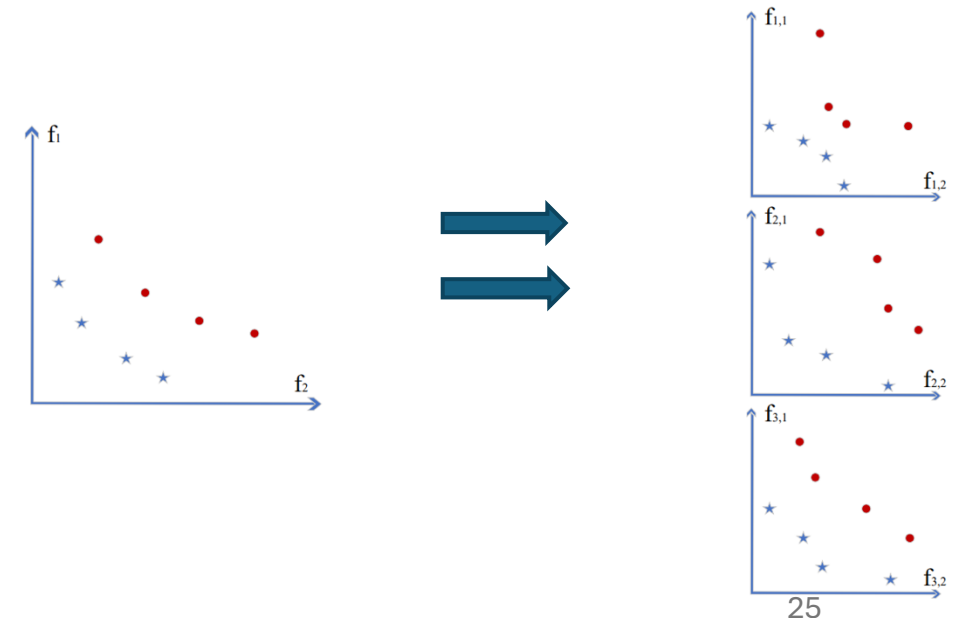
- Motivations:

- Can we utilize **more sampled data** from **other optimization problems**?
 - Can we enhance the **optimization results** with **other optimization problems**?
 - Can we solve multiple **few-shot** multiobjective optimization **together**?



Few-Shot Multiobjective Multitask Optimization!!!

$$\operatorname{argmin}_{x_m \in \mathcal{X}_m} F_m(x_m), m \in [M], \mathcal{C}_{m,\max} = \mathcal{C}$$



Why Few-Shot Multiobjective Multitask Optimization?

- Few-Shot Multiobjective Multitask Optimization:

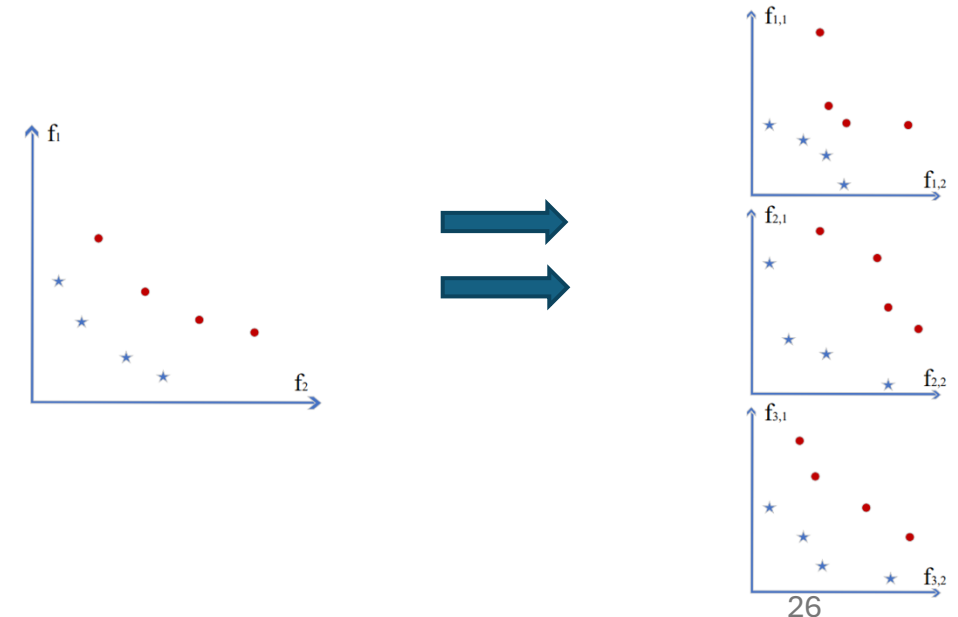
- $\operatorname{argmin}_{x_m \in \mathcal{X}_m} F_m(x_m), m \in [M], \mathcal{C}_{m, \max} = \mathcal{C}$
- exploit the **inter-task synergy** among the common multiobjective optimization problems
- to expedite the convergence performance
- and to generating high-quality Pareto solution set for each optimization task

$$\{\{(X_{k,i}^*, Y_{k,i}^*)\}_{i=1}^{M_k}\}_{k=1}^K \longrightarrow \{\psi_1^{-1}(\cdot), \dots, \psi_K^{-1}(\cdot)\}$$

- What is the **challenge** of Few-Shot Multiobjective Multitask Optimization?

- There is no prior work on Multitask Optimization simultaneously:

- Obtain high-quality multiobjective solutions.
- Generate high-quality inverse models.



Methodology

- Few-Shot Multiobjective Multitask Optimization:

- $\operatorname{argmin}_{x_m \in \mathcal{X}_m} F_m(x_m), m \in [M], \mathcal{C}_{m, \max} = \mathcal{C}$
- exploit the **inter-task synergy** among the common multiobjective optimization problems

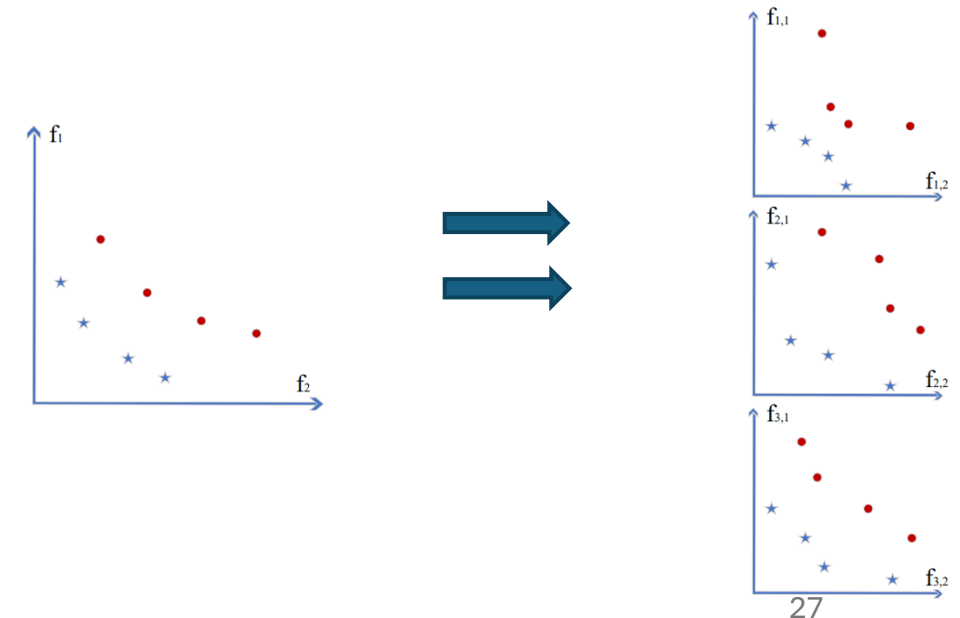
$$\{\{(X_{k,i}^*, Y_{k,i}^*)\}_{i=1}^{M_k}\}_{k=1}^K \longrightarrow \{\psi_1^{-1}(\cdot), \dots, \psi_K^{-1}(\cdot)\}$$

- Research Gap: No prior work on Multitask Optimization simultaneously:

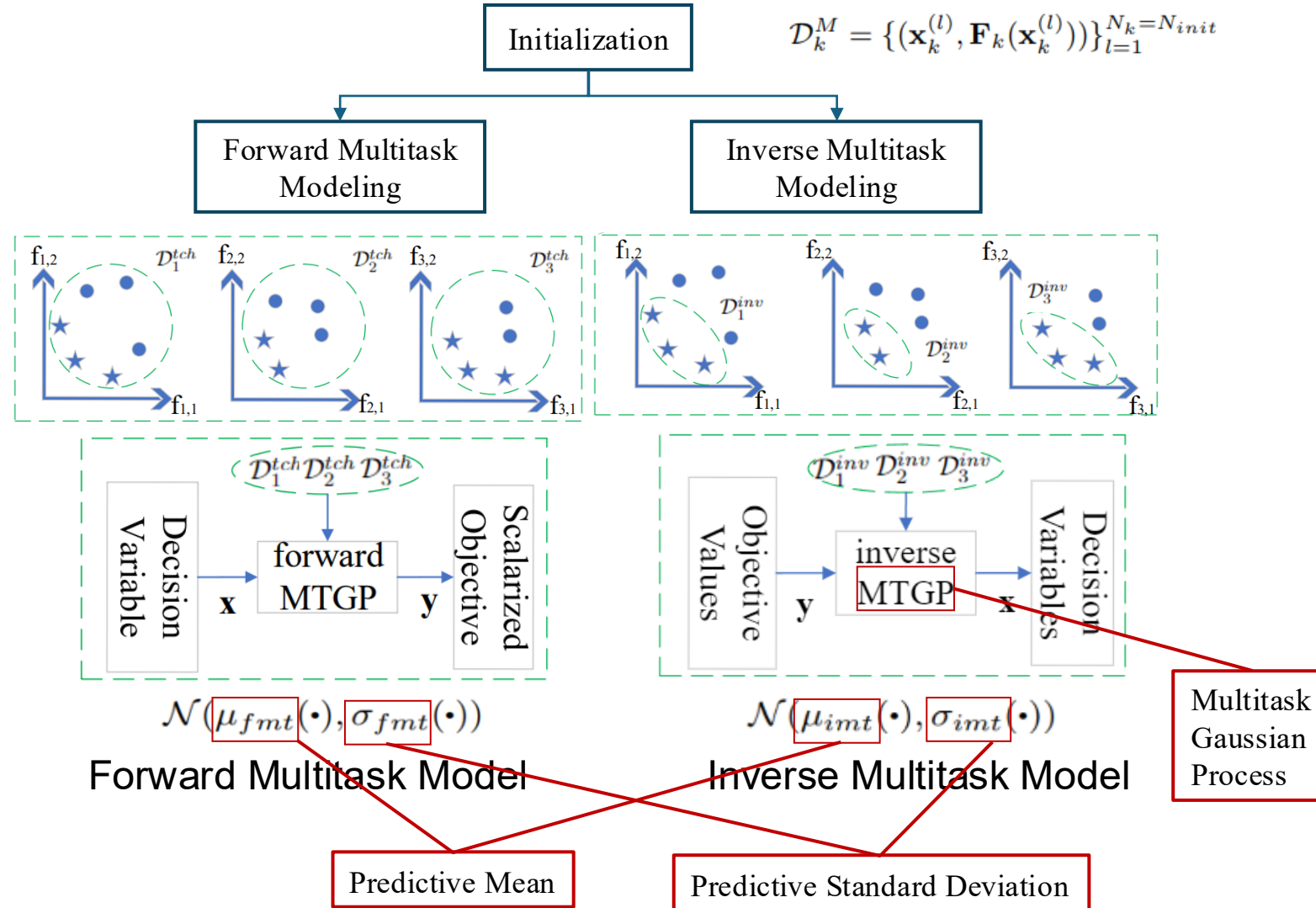
- Obtain high-quality multiobjective solutions.
- Generate high-quality inverse models.

- Contribution:

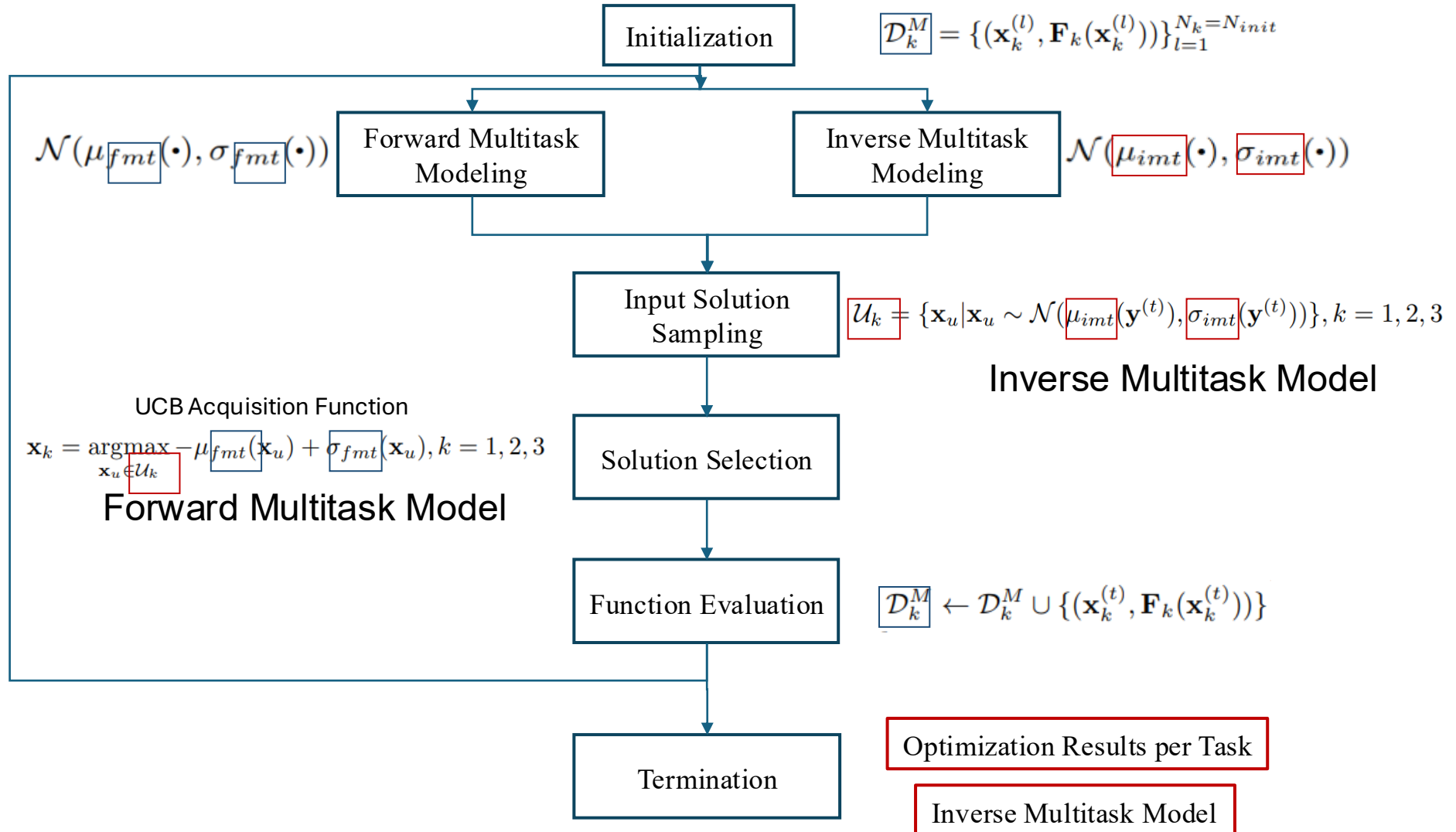
- Apply **forward-inverse model-based transfer**
- Forward model-based transfer enables better multiobjective solutions
- Inverse model-based transfer enables better inverse modeling performance



Forward-Inverse Model-based Transfer



Forward-Inverse Model-based Transfer



Few-Shot Multiobjective Multitask Optimization

- Research Gap: No prior work on Multitask Optimization simultaneously:

- Obtain high-quality multiobjective solutions.
- Generate high-quality inverse models.

$$\{\{(X_{k,i}^*, Y_{k,i}^*)\}_{i=1}^{M_k}\}_{k=1}^K \longrightarrow \{\psi_1^{-1}(\cdot), \dots, \psi_K^{-1}(\cdot)\}$$

- Contribution:

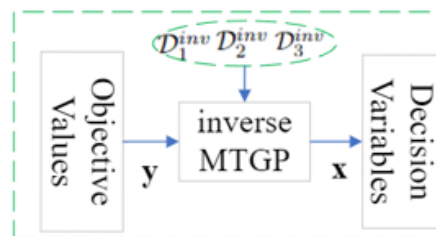
- Apply **forward-inverse model-based transfer**
- Forward model-based transfer enables better multiobjective solutions
- Inverse model-based transfer enables better inverse modeling performance

- Inverse model-based transfer?

- Utilizing the shared non-dominated solutions
- Enrich the inverse modeling training data

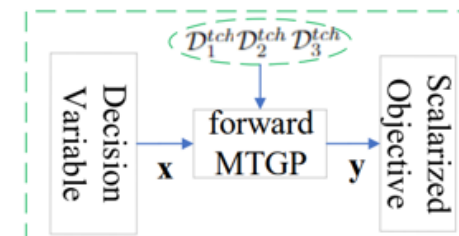
- Forward model-based transfer?

- Better quantify the inter-task relationship of function landscape
- Enrich the forward modeling training data
- Help obtain high-quality multiobjective solutions



$$\mathcal{N}(\mu_{imt}(\cdot), \sigma_{imt}(\cdot))$$

Inverse Multitask Model



$$\mathcal{N}(\mu_{fmt}(\cdot), \sigma_{fmt}(\cdot))$$

Forward Multitask Model

UCB Acquisition Function

$$\mathbf{x}_k = \underset{\mathbf{x}_u \in \mathcal{U}_k}{\operatorname{argmax}} -\mu_{fmt}(\mathbf{x}_u) + \sigma_{fmt}(\mathbf{x}_u), k = 1, 2, 3$$

Solution Selection

Experiments

Problems:

- Benchmark problems (modified from DTLZ problem sets^[1])
- Engineering optimization problems (truss design^[2])
- Hyperparameter Optimization Problems^[3]
 - (HPO-1)
 - XG-Boost on 3 machine learning classification tasks
 - Credit Approval^[4]
 - Medical Diagnosis^[4]
 - Speech Recognition^[4]
 - (HPO-2)
 - XG-Boost in 'gbtree' mode on speech recognition problem
 - XG-Boost in 'dart' mode on speech recognition problem

Table 3.1: Parameter settings of the mDTLZ benchmark problems. Each multiobjective problem contains three objectives and six-dimensional decision variables. In this table, three multiobjective multitask optimization problems are provided, each of which contains four tasks. The parameter s controls the biased search space, while other parameters control the shifted optima.

Task Set	Task Index	s	p_1	p_2	p_3	p_4
mDTLZ1	Task-1	0.35	0.50	0.50	0.50	0.50
	Task-2	0.30	0.25	0.50	0.75	1.00
	Task-3	0.25	0.125	0.25	0.375	0.50
	Task-4	0.20	0.083	0.167	0.25	0.33
mDTLZ2	Task-1	1.0	0.50	0.50	0.50	0.50
	Task-2	0.75	0.25	0.50	0.75	1.00
	Task-3	0.50	0.125	0.25	0.375	0.50
	Task-4	0.25	0.083	0.167	0.25	0.33
mDTLZ3	Task-1	0.20	0.50	0.50	0.50	0.50
	Task-2	0.18	0.25	0.50	0.75	1.00
	Task-3	0.16	0.125	0.25	0.375	0.50
	Task-4	0.14	0.083	0.167	0.25	0.33

$$\left\{ \begin{array}{l} f_1 = L(2x_1 + \sqrt{2}x_2 + \sqrt{x_3} + x_4), \\ f_2 = \frac{FL}{E} \left(\frac{2}{x_1} + \frac{2\sqrt{2}}{x_2} - \frac{2\sqrt{2}}{x_3} + \frac{2}{x_4} \right), \\ x_1, x_4 \in [a, 3a], \\ x_2, x_3 \in [\sqrt{2}a, 3a], \\ a = F/\sigma \end{array} \right.$$

Table 3.2: Parameter settings of the engineering design problem. This problem set contains three multiobjective optimization problems. Distinct parameter sets generate distinct loading cases for the four-bar truss design application. Each multiobjective optimization problem includes two objectives and four-dimensional decision variables.

Task Set	Task Index	F (kN)	σ (kN/cm ²)	L (cm)	E (kN/cm ²)
Four Truss Design	Task-1	10	10	200	2.0E5
	Task-2	10	8	200	1.8E5
	Task-3	8	5	200	1.5E5

[1] Deb, Kalyanmoy, et al. "Scalable multi-objective optimization test problems." Proceedings of the 2002 Congress on Evolutionary Computation. CEC'02 (Cat. No. 02TH8600). Vol. 1. IEEE, 2002.

[2] Tanabe, Ryoji, and Hisao Ishibuchi. "An easy-to-use real-world multi-objective optimization problem suite." Applied Soft Computing 89 (2020): 106078.

[3] Pfisterer, Florian, et al. "Yahpo gym-an efficient multi-objective multi-fidelity benchmark for hyperparameter optimization." International Conference on Automated Machine Learning. PMLR, 2022.

[4] Vanschoren, Joaquin, et al. "OpenML: networked science in machine learning." ACM SIGKDD Explorations Newsletter 15.2 (2014): 49-60.

Experiments

- Methods:

- ParEGO^[1]
 - A baseline few-shot multiobjective optimizer
 - Inverse Model? No
 - Multitask? No
- PSL-MOBO^[2]
 - Deemed as one of state-of-the-art few-shot multiobjective optimizers
 - Inverse Model? Yes
 - Multitask? No
- UCB-Tch^[3]
 - Deemed as one of state-of-the-art few-shot multiobjective optimizers
 - Inverse Model? No
 - Multitask? No
- Ours
 - Inverse Model? Yes
 - Multitask? Yes

[1] Knowles, Joshua. "ParEGO: A hybrid algorithm with on-line landscape approximation for expensive multiobjective optimization problems." IEEE transactions on evolutionary computation 10.1 (2006): 50-66.

[2] Lin, Xi, et al. "Pareto set learning for expensive multi-objective optimization." Advances in Neural Information Processing Systems 35 (2022): 19231-19247.

[3] Paria, Biswajit, Kirthevasan Kandasamy, and Barnabás Póczos. "A flexible framework for multi-objective bayesian optimization using random scalarizations." Uncertainty in Artificial Intelligence. PMLR, 2020.

Results

- Evaluation Metrics:

- Quality of approximated Pareto solution sets
- Accuracy of the inverse model

- Forward model-based transfer?
 - Better quantify the inter-task relationship of function landscape
 - Enrich the forward modeling training data
 - Help obtain high-quality multiobjective solutions

TABLE S-IV
HYPERVOLUME RESULTS ON BENCHMARK PROBLEMS AND REAL-WORLD PROBLEMS.

Problems	Tasks	ParEGO	PSL-MOBO	UCB-Tch	F-invTrEMO
mDTLZ-1	Task-1	1.6083 (0.0193) ≈	1.5188 (0.0883) –	1.6001 (0.0352) ≈	1.6111 (0.0154)
	Task-2	1.4391 (0.1746) ≈	1.3649 (0.1733) –	1.4868 (0.1028) ≈	1.5139 (0.1529)
	Task-3	1.5439 (0.0542) –	1.3545 (0.1080) –	1.5493 (0.0480) –	1.5960 (0.0291)
	Task-4	1.4871 (0.0761) –	1.2077 (0.0928) –	1.4420 (0.1018) –	1.5632 (0.1189)
mDTLZ-2	Task-1	0.8030 (0.0287) –	1.0204 (0.0156) ≈	1.0410 (0.0106) +	1.0273 (0.0154)
	Task-2	0.6856 (0.0403) –	0.8929 (0.0663) –	0.9696 (0.0243) ≈	0.9764 (0.0281)
	Task-3	0.6888 (0.0354) –	0.7174 (0.0315) –	0.9293 (0.0409) –	0.9901 (0.0267)
	Task-4	0.5574 (0.0550) –	0.4588 (0.0480) –	0.6688 (0.0822) ≈	0.7786 (0.2547)
mDTLZ-3	Task-1	0.0997 (0.0340) –	0.0647 (0.0224) –	0.0784 (0.0370) –	0.1510 (0.0255)
	Task-2	0.0705 (0.0325) –	0.0752 (0.0153) –	0.0494 (0.0261) –	0.1394 (0.0497)
	Task-3	0.0831 (0.0439) –	0.0529 (0.0223) –	0.0568 (0.0438) –	0.1524 (0.0268)
	Task-4	0.0623 (0.0156) –	0.0618 (0.0202) –	0.0430 (0.0347) –	0.1272 (0.0365)
EO	Task-1	0.7532 (0.0026) –	0.7144 (0.0133) –	0.7501 (0.0043) –	0.7589 (0.0036)
	Task-2	0.7546 (0.0043) –	0.7297 (0.0155) –	0.7534 (0.0041) –	0.7607 (0.0033)
	Task-3	0.7579 (0.0030) –	0.7286 (0.0130) –	0.7546 (0.0046) –	0.7632 (0.0032)
HPO-1	Task-1	1.4996 (0.0054) –	1.5073 (0.0048) ≈	1.5054 (0.0054) ≈	1.5052 (0.0035)
	Task-2	1.6114 (0.0052) –	1.6159 (0.0025) –	1.6166 (0.0089) –	1.6210 (0.0046)
	Task-3	1.5377 (0.0052) –	1.5311 (0.0080) –	1.5302 (0.0146) –	1.5411 (0.0054)
HPO-2	Task-1	1.3433 (0.0060) –	1.3463 (0.0049) –	1.3477 (0.0057) –	1.3514 (0.0025)
	Task-2	1.3466 (0.0036) –	1.3468 (0.0023) –	1.3442 (0.0060) –	1.3503 (0.0029)

Results

- Evaluation Metrics:
 - Quality of approximated Pareto solution sets
 - Accuracy of the inverse model
 - Only ours and PSL-MOBO can generate inverse models

Table 2: RMSE comparison results on benchmark problems and real-world problems. Numbers with indicators (+), (−) and ($\approx$) imply that the compared algorithm is better than, worse than, or similar to F-invTrEMO at 95% confidence level as per the Wilcoxon signed-rank test.

Problems	Tasks	PSL-MOBO	F-invTrEMO
mDTLZ-1	Task-1	0.3021 (0.0744) −	0.0721 (0.0173)
	Task-2	0.3828 (0.0770) $\approx$	0.4250 (0.2263)
	Task-3	0.3790 (0.0837) −	0.1592 (0.0780)
	Task-4	0.4019 (0.0787) −	0.3568 (0.1691)
mDTLZ-2	Task-1	0.3501 (0.1321) −	0.0679 (0.0295)
	Task-2	0.2997 (0.0146) −	0.1436 (0.0219)
	Task-3	0.7010 (0.1819) −	0.1700 (0.0454)
	Task-4	0.9420 (0.1361) −	0.3248 (0.1767)
mDTLZ-3	Task-1	1.0219 (0.0310) −	0.6946 (0.0392)
	Task-2	1.0430 (0.0429) −	0.7412 (0.0325)
	Task-3	1.0562 (0.0457) −	0.7830 (0.0361)
	Task-4	1.1166 (0.0747) −	0.8301 (0.0423)
EO	Task-1	0.2211 (0.0242) $\approx$	0.2173 (0.0054)
	Task-2	0.2290 (0.0261) −	0.2144 (0.0067)
	Task-3	0.2225 (0.0372) $\approx$	0.2046 (0.0053)
HPO-1	Task-1	0.1818 (0.1143) −	0.0943 (0.0721)
	Task-2	0.1663 (0.1558) −	0.0626 (0.0125)
	Task-3	0.1881 (0.0545) −	0.1273 (0.0411)
HPO-2	Task-1	0.1855 (0.1075) −	0.0811 (0.0476)
	Task-2	0.1572 (0.1612) −	0.0694 (0.0260)

- Inverse model-based transfer?
 - Utilizing the shared non-dominated solutions
 - Enrich the inverse modeling training data

Results (Ablation)

- Evaluation Metrics:
 - Quality of approximated Pareto solution sets
 - Accuracy of the inverse model

Table 3.7: IGD and RMSE results on benchmark problems for F-invTrEMO with and without forward multitask GP (invTrEMO). Numbers (means and standard deviations) with indicators (+), (−) and ($\approx$) imply that the compared algorithm is better than, worse than, or similar to F-invTrEMO at 95% confidence level as per the Wilcoxon signed-rank test.

Problems	Tasks	IGD		RMSE	
		invTrEMO	F-invTrEMO	invTrEMO	F-invTrEMO
mDTLZ-1	Task-1	0.0820 (0.0126) −	0.0611 (0.0051)	0.1174 (0.0358) −	0.0721 (0.0173)
	Task-2	0.1264 (0.0699) $\approx$	0.1528 (0.1200)	0.4857 (0.2244) $\approx$	0.4250 (0.2263)
	Task-3	0.1203 (0.0301) −	0.0731 (0.0099)	0.3181 (0.1851) −	0.1592 (0.0780)
	Task-4	0.1457 (0.0427) $\approx$	0.1182 (0.0718)	0.4911 (0.2010) −	0.3568 (0.1691)
mDTLZ-2	Task-1	0.1157 (0.0074) $\approx$	0.1135 (0.0083)	0.0637 (0.0244) $\approx$	0.0679 (0.0295)
	Task-2	0.1284 (0.0118) +	0.1489 (0.0174)	0.1466 (0.0484) $\approx$	0.1436 (0.0219)
	Task-3	0.1350 (0.0120) $\approx$	0.1328 (0.0150)	0.1556 (0.0273) $\approx$	0.1700 (0.0454)
	Task-4	0.2192 (0.1015) −	0.1439 (0.0154)	0.3360 (0.1605) $\approx$	0.3248 (0.1767)
mDTLZ-3	Task-1	0.3090 (0.1115) −	0.2726 (0.0277)	0.7355 (0.0633) −	0.6946 (0.0392)
	Task-2	0.4589 (0.1682) −	0.3059 (0.0399)	0.8288 (0.1004) −	0.7412 (0.0325)
	Task-3	0.3103 (0.0743) −	0.2833 (0.0356)	0.7837 (0.0650) $\approx$	0.7830 (0.0361)
	Task-4	0.3941 (0.1475) −	0.3241 (0.0305)	0.8403 (0.0803) $\approx$	0.8301 (0.0423)

Only inverse multitask model

Inverse and forward multitask model

Extension: Multitask Optimization via Sequential Transfer

- Few-Shot Multiobjective Multitask Optimizer:

- Each pair of tasks act like both the source task and target task

Not all the source-target pairs can benefit from knowledge transfer.

- Allow each task conduct same/similar evaluations each iteration

Treating all tasks equally may lead to insufficient resource utilization.

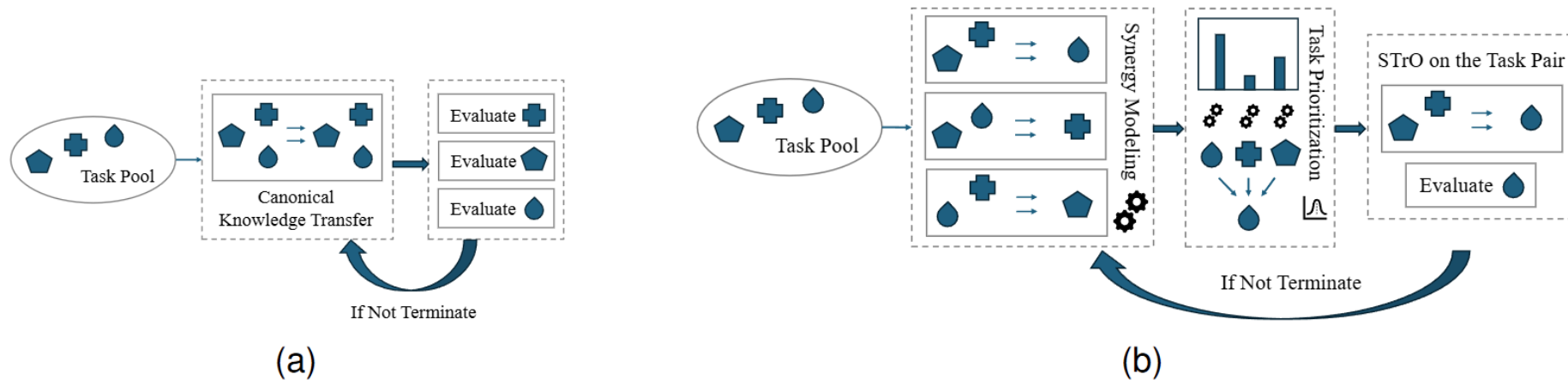
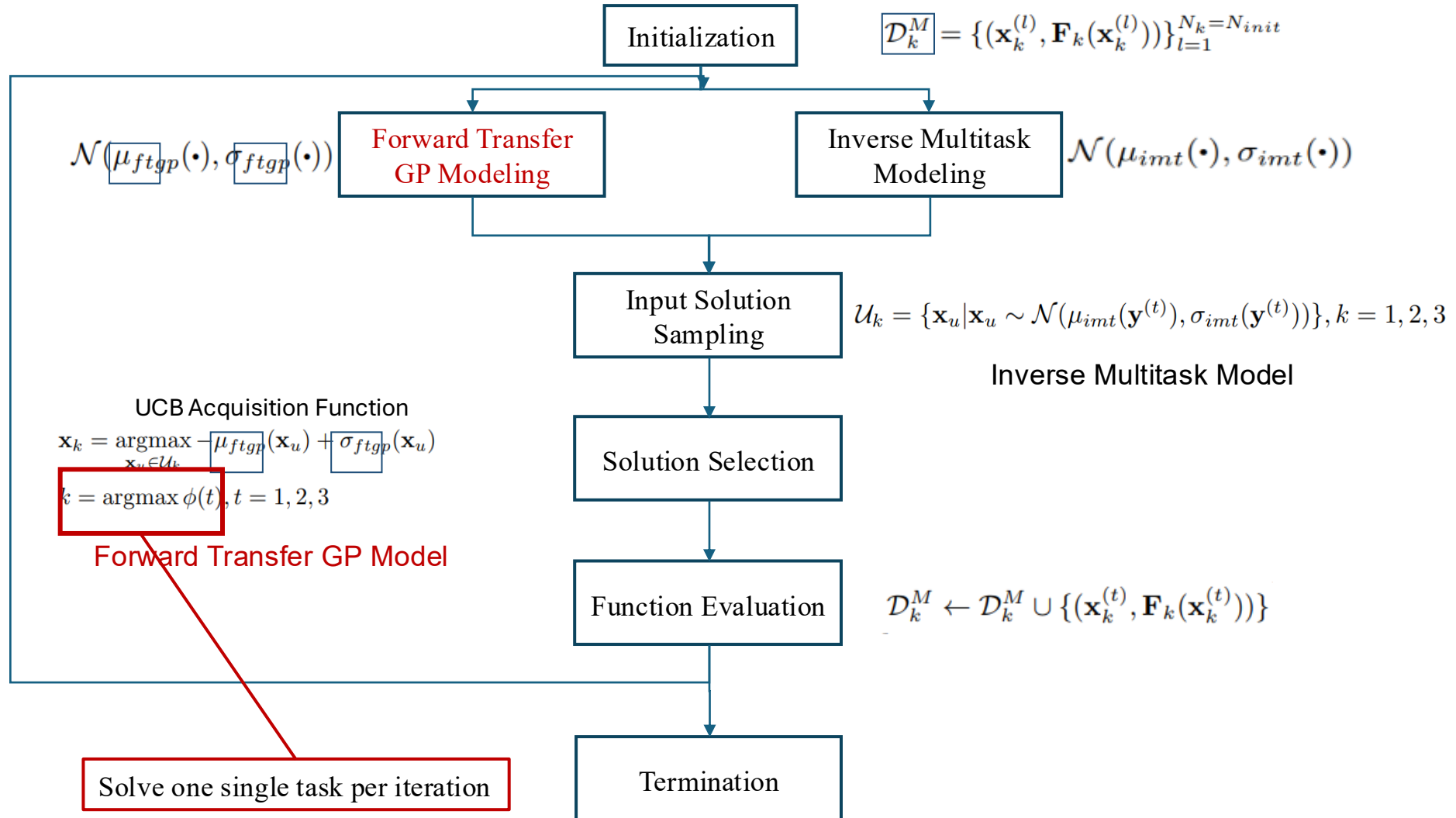


Fig. 1. Workflow of distinct multitask optimization frameworks. (a) Standard multitask optimization (tasks evaluated evenly) (b) The proposed multitask optimization with iterative sequential transfer optimization (tasks evaluated selectively).

Solve one target task per iteration rather than solve all the tasks simultaneously.

Extension: Multitask Optimization via Sequential Transfer



Extension: Multitask Optimization via Sequential Transfer

- Forward Transfer GP Model: $\mathcal{N}(\mu_{ftgp}(\cdot), \sigma_{ftgp}(\cdot))$
 - For source tasks 1, 2, ..., K and a target task. We build K transfer GP models:

$$\tilde{k}(\mathbf{x}, \mathbf{x}') = \begin{cases} \lambda k(\mathbf{x}, \mathbf{x}'), & \mathbf{x} \in \mathbf{X}_S \text{ \& } \mathbf{x}' \in \mathbf{X}_T \\ & \text{or } \mathbf{x} \in \mathbf{X}_T \text{ \& } \mathbf{x}' \in \mathbf{X}_S \\ k(\mathbf{x}, \mathbf{x}'), & \text{otherwise.} \end{cases}$$

- $\lambda_{i,j} \in [-1,1]$ is a learnt kernel hyperparameter that can capture positive/negative correlations between source i and target j

$$\phi(t) = \min |\lambda_{i,t}|, \quad i = 1, \dots, K$$

- We select the target task with the largest function $\phi(t)$
- Why?

Detecting negative transfer

Less likely to occur negative transfer

- GP-UCB optimization regret can be bounded by $\mathbf{I}([y^{(1)}, y^{(2)}, \dots, y^{(N)}]; \mathbf{f}_T) = \frac{1}{2} \log |I + \sigma_n^{-2} K_{TT}|$
- GP-UCB with transfer, the regret can be bounded by

$$\begin{aligned} & \mathbf{I}([y_T^{(1)}, \dots, y_T^{(N_T)}]; \mathbf{f}_T | \mathcal{D}_S, \lambda, M) \\ &= \frac{1}{2} \log |I + \sigma_T^{-2} (K_{TT} - \lambda^2 K'_{TS} (K_{SS} + \sigma_S^2 I)^{-1} K'_{ST})| \end{aligned}$$

- Larger $|\lambda_{i,t}|$ stand for a tighter regret bound, meaning a more accurate GP model can be constructed^[1].

Extension: Multitask Optimization via Sequential Transfer

Results

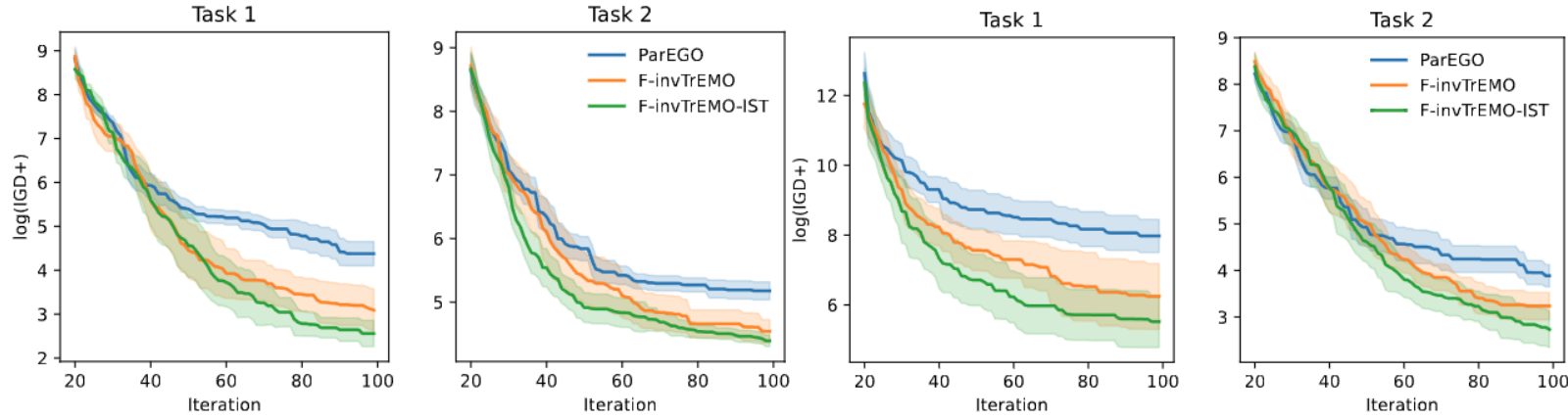


TABLE I

IGD+ COMPARISON RESULTS ON MULTITASK OPTIMIZATION BENCHMARK FOR PAREGO, F-INVTRIMO, AND F-INVTRIMO-IST

Problems	Tasks	ParEGO	F-invTrEMO	F-invTrEMO-IST
CIHS	Task-1	1.516E+02 (1.545E+01) +	8.061E+01 (1.534E+01) +	7.156E+01 (1.142E+01)
	Task-2	4.877E+00 (3.150E-01) +	3.557E+00 (2.329E-01) ≈	3.389E+00 (1.925E-01)
CIMS	Task-1	2.523E+02 (2.057E+01) +	2.086E+02 (2.658E+01) +	1.954E+02 (4.069E+01)
	Task-2	4.174E+00 (2.983E-01) +	3.819E+00 (2.859E-01) +	3.606E+00 (1.799E-01)
CILS	Task-1	4.564E+01 (2.021E+00) +	4.435E+01 (1.592E+00) +	4.113E+01 (1.601E+00)
	Task-2	3.050E-01 (1.040E-02) -	7.938E-01 (5.305E-02) +	6.856E-01 (4.870E-02)
PIHS	Task-1	8.728E+01 (9.112E+00) +	3.234E+01 (8.086E+00) +	1.399E+01 (1.978E+00)
	Task-2	1.841E+02 (1.678E+01) +	9.913E+01 (1.033E+01) +	8.109E+01 (3.761E+00)
PIMS	Task-1	2.678E+00 (1.539E-01) +	4.324E-01 (3.790E-02) -	4.784E-01 (5.205E-02)
	Task-2	3.736E+02 (1.269E+01) +	3.503E+02 (5.258E+00) ≈	3.407E+02 (4.165E+00)
PILS	Task-1	1.008E+00 (1.390E-02) ≈	1.029E+00 (1.165E-02) ≈	1.041E+00 (9.500E-03)
	Task-2	1.423E+01 (6.948E-01) +	1.258E+01 (8.887E-01) +	1.053E+01 (6.600E-01)
NIHS	Task-1	3.174E+03 (6.287E+02) +	1.651E+03 (5.881E+02) +	5.934E+02 (2.174E+02)
	Task-2	5.820E+01 (4.852E+00) +	2.834E+01 (3.824E+00) +	1.903E+01 (4.056E+00)

TABLE II
IGD+ COMPARISON RESULTS ON MULTITASK HYPERPARAMETER TUNING PROBLEMS FOR F-INVTRIMO AND F-INVTRIMO-IST

Problems	Tasks	F-invTrEMO	F-invTrEMO-IST
HPO-1	Task-1	0.0502 (0.0028) +	0.0455 (0.0021)
	Task-2	0.0439 (0.0020) ≈	0.0423 (0.0014)
HPO-2	Task-1	0.0325 (0.0017) +	0.0303 (0.0031)
	Task-2	0.0282 (0.0019) ≈	0.0279 (0.0014)
	Task-3	0.0421 (0.0030) +	0.0392 (0.0023)

TABLE S-I

IGD+ COMPARISON RESULTS ON MULTITASK OPTIMIZATION BENCHMARK FOR PAREGO, AMTEA, AND AMTEA-IST

Problems	Tasks	ParEGO	AMTEA	AMTEA-IST
CIHS	Task-1	8.412E+01 (8.859E+00) +	6.392E+01 (5.758E+00) ≈	6.314E+01 (4.842E+00)
	Task-2	3.673E+00 (1.373E-01) ≈	3.202E+00 (1.530E-01) -	3.663E+00 (2.081E-01)
CIMS	Task-1	1.251E+02 (6.850E+00) +	1.026E+02 (1.079E+01) +	8.299E+01 (8.457E+00)
	Task-2	2.668E+00 (9.690E-02) -	2.554E+00 (1.140E-01) -	2.870E+00 (1.923E-01)
CILS	Task-1	3.400E+01 (1.934E+00) -	4.209E+01 (1.494E+00) +	3.810E+01 (1.712E+00)
	Task-2	2.243E-01 (5.650E-03) +	2.136E-01 (7.850E-03) +	1.892E-01 (8.900E-03)
PIHS	Task-1	6.839E+01 (1.157E+01) +	5.505E+01 (6.409E+00) +	4.675E+01 (8.927E+00)
	Task-2	1.454E+02 (7.436E+00) +	1.356E+02 (9.290E+00) ≈	1.295E+02 (8.145E+00)
PIMS	Task-1	2.158E+00 (2.162E-01) +	8.786E-01 (1.249E-01) ≈	8.410E-01 (4.580E-02)
	Task-2	3.377E+02 (4.546E+00) ≈	3.355E+02 (4.538E+00) ≈	3.335E+02 (4.739E+00)
PILS	Task-1	1.016E+00 (6.050E-03) +	9.833E-01 (9.000E-03) ≈	9.482E-01 (1.635E-02)
	Task-2	9.252E+00 (4.003E-01) +	7.599E+00 (3.862E-01) ≈	7.877E+00 (5.489E-01)
NIHS	Task-1	9.446E+02 (1.847E+02) +	4.015E+02 (7.916E+01) +	3.457E+02 (6.560E+01)
	Task-2	3.584E+01 (2.767E+00) +	3.466E+01 (3.816E+00) ≈	3.344E+01 (1.626E+00)

Outline

- ❑ Research Topic Introduction
 - ❑ Multiobjective Optimization
 - ❑ Few-Shot Challenges
 - ❑ Multitask Optimization
 - ❑ From Multitask Optimization to Parametric Multitask Optimization
- ❑ Problem 1: Finite Few-Shot Single-Objective Multitask Optimization
 - In-Context Guidance: Learning Inter-Task Synergies via Numerical Foundational Models for Few-Shot Multitask Optimization^[1]
- ❑ Problem 2: Finite Few-Shot Multiobjective Multitask Optimization
 - Bayesian Forward-Inverse Transfer for Multiobjective Optimization^[2]
 - Solving Few-Shot Multiobjective Multitask Optimization via Iterative Sequential Transfer^[3]
- ❑ Problem 3: Parametric Single-Objective Multitask Optimization
 - (θ_l, θ_u) -Parametric Multi-Task Optimization: Joint Search in Solution and Infinite Task Spaces^[4]
- ❑ Problem 4: Parametric Multiobjective Multitask Optimization
 - Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling^[5]
- ❑ Conclusions and Potential Future Work

[1] Tingyang Wei, et al. “In-Context Guidance: Learning Inter-Task Synergies via Numerical Foundational Models for Few-Shot Multitask Optimization.” In submission to IEEE Transactions on Evolutionary Computation (2026)

[2] Tingyang Wei, et al. “Bayesian forward-inverse transfer for multiobjective optimization.” International Conference on Parallel Problem Solving from Nature. Cham: Springer Nature Switzerland, 2024.

[3] Tingyang Wei, et al. “Solving Few-Shot Multiobjective Multitask Optimization via Iterative Sequential Transfer.” 2026 IEEE Congress on Evolutionary Computation (CEC).

[4] Tingyang Wei, Jiao Liu, Abhishek Gupta, Puay Siew Tan, Yew-Soon Ong, “ (θ_l, θ_u) -Parametric Multi-Task Optimization: Joint Search in Solution and Infinite Task Spaces” IEEE Transactions on Evolutionary Computation (2025).

[5] Tingyang Wei, Jiao Liu, Abhishek Gupta, Chin Chun Ooi, Puay Siew Tan, Yew-Soon Ong, “Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling”, *IJCAI* 2026.

Parametric Multitask Optimization

▪ Multitask Optimization:

- $\operatorname{argmin}_{x_m \in \mathcal{X}_m} F_m(x_m), m \in [M]$
- Finite Tasks

Assuming M problems share certain similarities, we use their **solution distributions and surrogate models** as transferable knowledge to accelerate convergence.

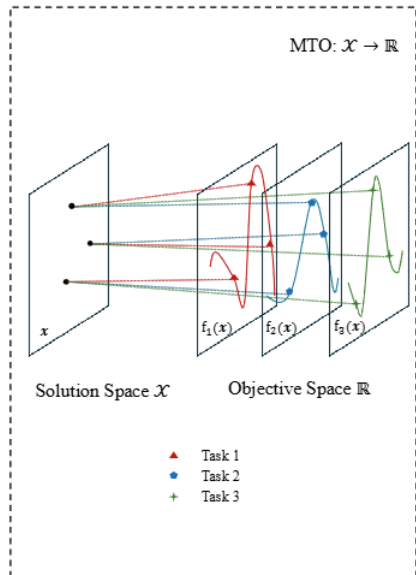
▪ Parametric Multitask Optimization:

- $\operatorname{argmin}_{x \in \mathcal{X}} F(x, \theta), \forall \theta \in \Theta$
- Infinite Tasks

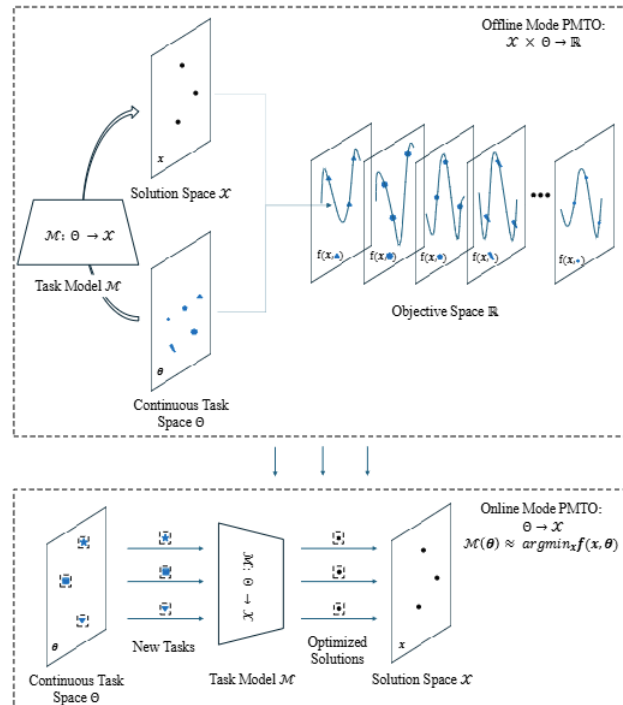
We assume access to **an external task space Θ** and investigate how leveraging **task parameters θ as side information** can enhance knowledge transfer and accelerate optimization convergence.

Cross-Task
Information as
Knowledge

Explicit
Parameter as
Knowledge



(a) Multi-task Optimization



(b) Parametric Multi-task Optimization

Tingyang Wei, Jiao Liu, Abhishek Gupta, Puay Siew Tan, Yew-Soon Ong, “ (θ_l, θ_u) -Parametric Multi-Task Optimization: Joint Search in Solution and Infinite Task Spaces” IEEE Transactions on Evolutionary Computation (2025).

$$\operatorname{argmin}_{x \in \mathcal{X}} f(x, \theta), \forall \theta \in \Theta$$

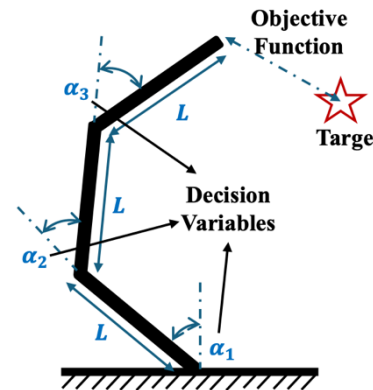
Why Parametric Multitask Optimization?

- Opportunities:

- Explicit task parameters θ provide a structured way to quantify task similarity, contributing to faster optimization convergence and more efficient knowledge transfer.

- Application Example:

- In robotic control design, θ can represent different morphologies. Optimizing control policy x for diverse θ allows robots to unexpected environments.
- Similar θ entails similar x
- Backing up x -s for diverse θ can enable adaptation



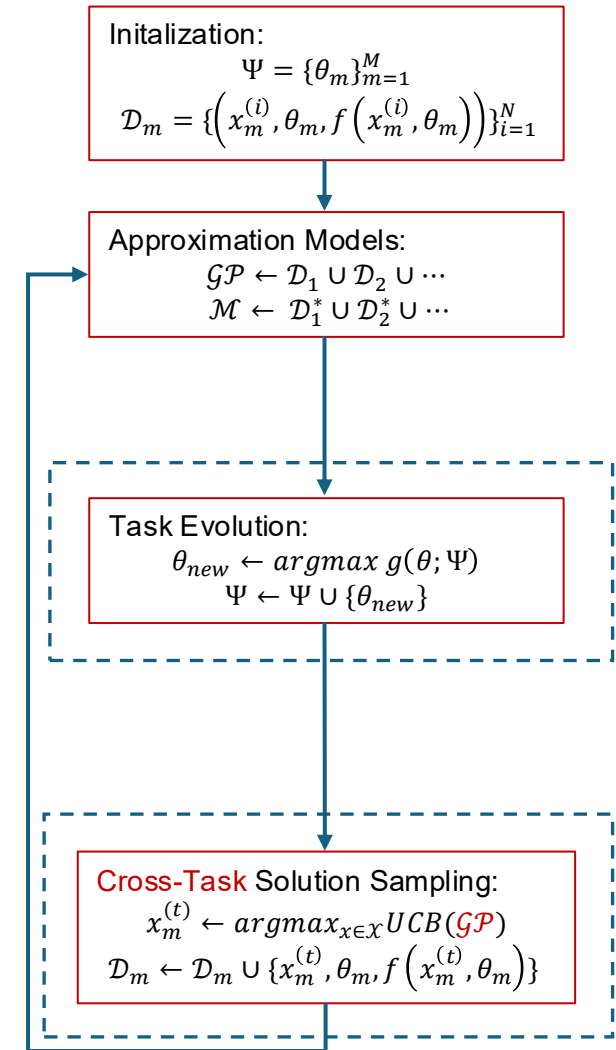
$$x: (\alpha_1, \alpha_2, \alpha_3)$$

$$\theta: (L, \alpha_{max})$$

$$\forall \theta \in \Theta, \mathcal{M}(\theta) = \operatorname{argmin}_{x \in \mathcal{X}} f(x, \theta)$$

Methodology

- $\forall \theta \in \Theta, \mathcal{M}(\theta) = \operatorname{argmin}_{x \in \mathcal{X}} f(x, \theta)$
- Surrogate Models:
 - (Forward Model) $\mathcal{GP}: \mathcal{X} \times \Theta \rightarrow \mathbb{R}$
 - (Inverse Model) $\mathcal{M}: \Theta \rightarrow \mathcal{X}$
- Task Evolution:
 - (Find representative task parameter)
 - $g(\theta; \Psi) = \det Q([\theta_1, \dots, \theta_M, \theta])$
 - (Optimal coverage of the task space)
- Cross-Task Sampling:
 - $x_m^{(t)} \leftarrow \operatorname{argmax}_{x \in \mathcal{X}} -\mu(x, \theta_m) + \beta \sigma(x, \theta_m)$
 - Forward Model contains cross-task information:
 - $\mathcal{GP}(\mu(\cdot, \theta), \sigma^2(\cdot, \theta))$



Performance Metrics

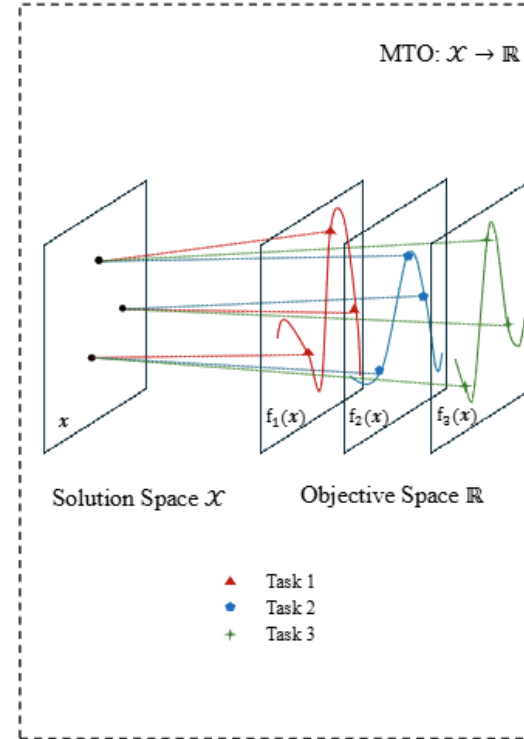
- $\forall \theta \in \Theta, \mathcal{M}(\theta) = \operatorname{argmin}_{x \in \mathcal{X}} f(x, \theta)$

- Optimization Results

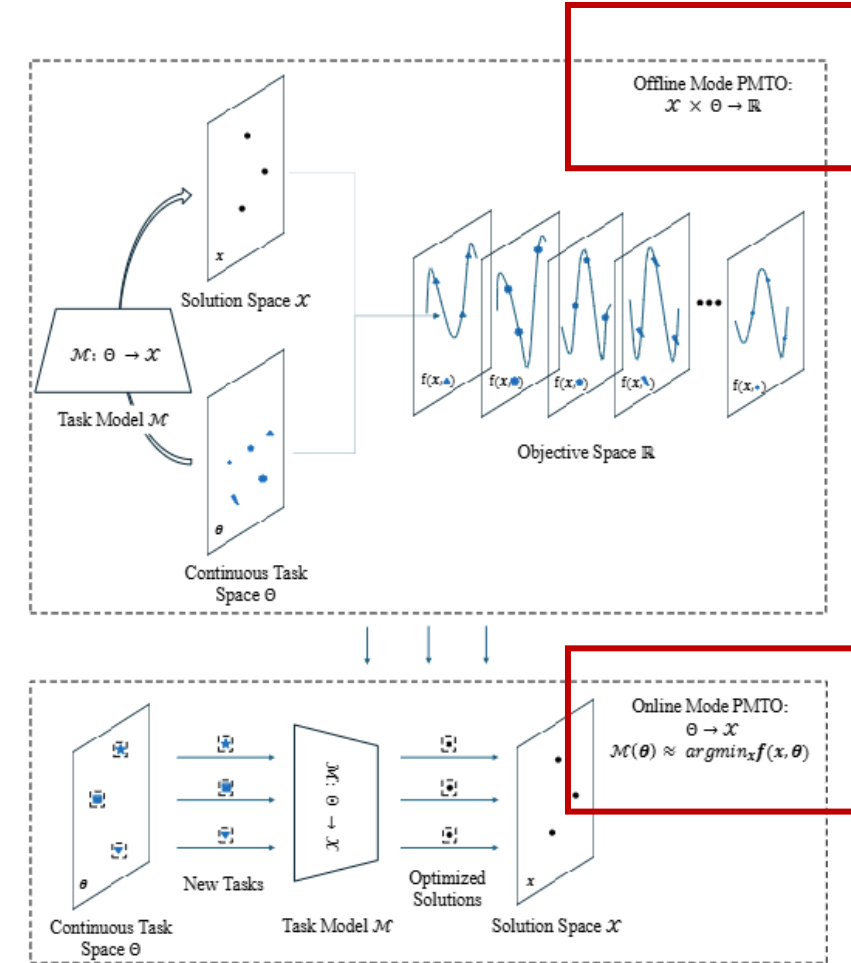
- (Forward Model) $\mathcal{GP}: \mathcal{X} \times \Theta \rightarrow \mathbb{R}$

- Inverse Model Performance

- (Inverse Model) $\mathcal{M}: \Theta \rightarrow \mathcal{X}$
 - $\mathcal{M}(\theta) \approx \operatorname{argmin}_{x \in \mathcal{X}} f(x, \theta)$



(a) Multi-task Optimization



(b) Parametric Multi-task Optimization

Results

- $\operatorname{argmin}_{x \in \mathcal{X}} f(x, \theta), \forall \theta \in \Theta$

- Good Optimization Results

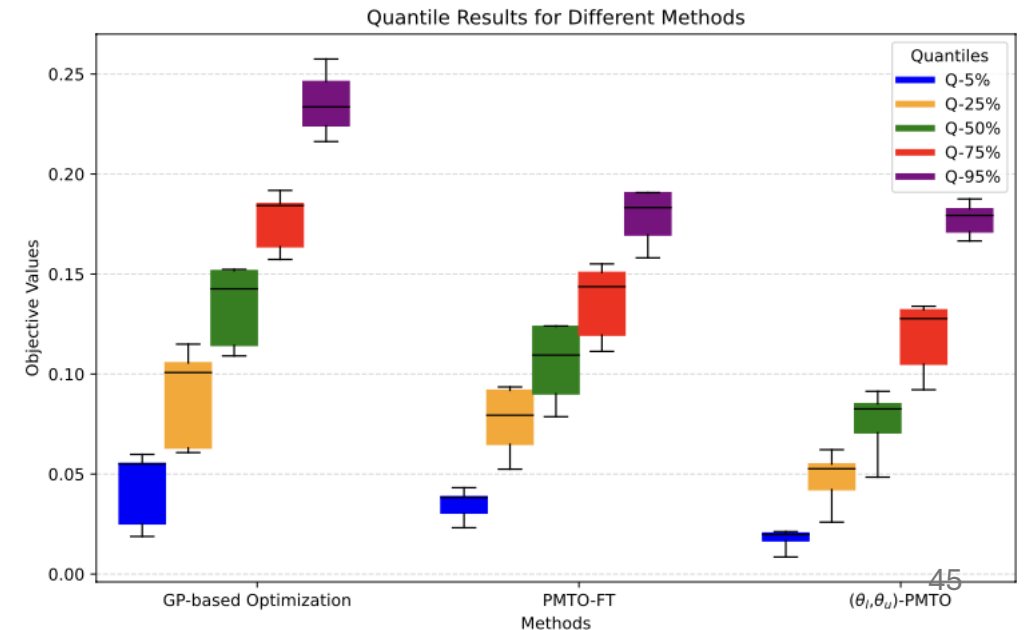
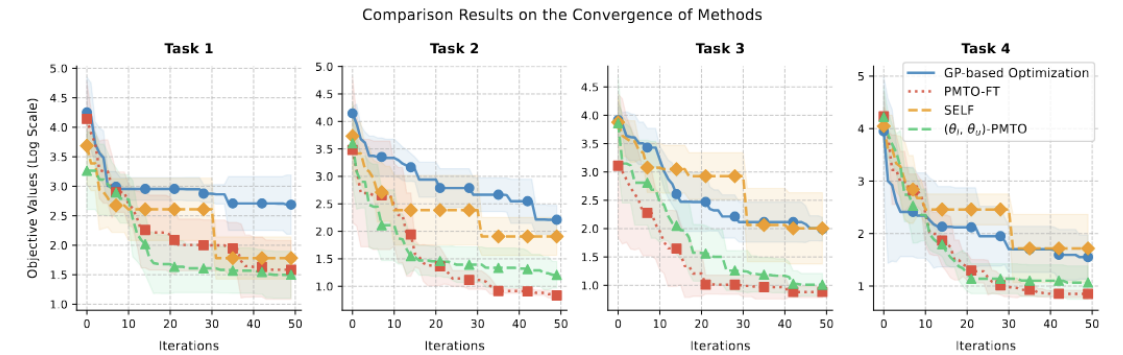
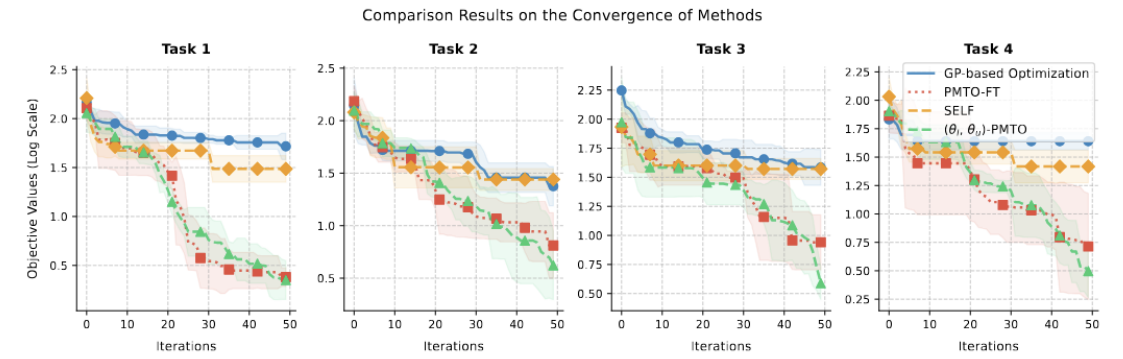
- Sample multiple tasks
- Check out the convergence
 - $f(\cdot, \theta_m), m \in [M]$
- Theoretical Analysis of faster convergence

Theorem. Let the kernel functions used in PMTO-FT and the independent strategy satisfy $\kappa_{pmt}((\mathbf{x}, \theta), (\mathbf{x}', \theta)) = \kappa_{ind}(\mathbf{x}, \mathbf{x}')$. Then, the MIG in the independent strategy, denoted as $\gamma_{T,m}^{ind}$, and the MIG in the PMTO-FT, denoted as $\gamma_{T,m}$, satisfy $\gamma_{T,m} \leq \gamma_{T,m}^{ind}, \forall m \in [M], \forall T \geq 1$.

- $\forall \theta \in \Theta, \mathcal{M}(\theta) = \operatorname{argmin}_{x \in \mathcal{X}} f(x, \theta)$

- Good Inverse Model Predictive Results

- Sample multiple (seen and unseen) tasks θ
- Check out the solution quality of the model output
 - $f(x', \theta), x' = \mathcal{M}(\theta)$



Detailed Results

TABLE I

COMPARATIVE RESULTS OF THE TASK MODEL’S ONLINE PERFORMANCE. THE RESULTS INCLUDE SINGLE-TASK BASELINE METHOD, MTO METHOD, AND PMTO VARIANTS (WITH OR WITHOUT TASK EVOLUTION) ON SYNTHETIC TEST PROBLEMS. $U = 20$ INDEPENDENT TRIALS ARE CONSIDERED.

Problems	Quantile	GP-based Optimization	SELF	PMTO-FT	(θ_t, θ_u) -PMTO-RT	(θ_t, θ_u) -PMTO
Sphere-I	5%	3.4605e-02 (2.2632e-02) $\approx$	3.5375e-02 (1.6824e-02) $\approx$	3.3402e-02 (2.6492e-02) $\approx$	8.7646e-02 (6.5269e-02) $-$	4.3264e-02 (1.6935e-02)
	25%	1.2633e-01 (7.6585e-02) $\approx$	1.5640e-01 (8.1991e-02) $\approx$	1.2448e-01 (1.1445e-01) $\approx$	2.7083e-01 (2.2507e-01) $-$	1.2737e-01 (5.0277e-02)
	50%	2.9818e-01 (1.5592e-01) $\approx$	4.3039e-01 (1.7511e-01) $-$	3.1267e-01 (2.8268e-01) $\approx$	5.8800e-01 (5.7331e-01) $-$	2.4507e-01 (1.0836e-01)
	75%	8.9086e-01 (3.0498e-01) $-$	1.0883e+00 (3.1372e-01) $-$	6.5575e-01 (3.6466e-01) $-$	1.2854e+00 (1.2831e+00) $-$	4.3753e-01 (2.3145e-01)
	95%	3.4692e+00 (6.0544e-01) $-$	3.6327e+00 (4.5250e-01) $-$	2.5749e+00 (1.0022e+00) $-$	3.3264e+00 (2.8819e+00) $-$	8.3653e-01 (4.1045e-01)
Sphere-II	5%	1.2980e+00 (8.1092e-02) $-$	1.3832e+00 (2.1743e-02) $-$	1.3668e+00 (4.8590e-02) $-$	1.0247e+00 (2.8961e-01) $-$	4.2056e-01 (2.5906e-01)
	25%	2.2877e+00 (8.6464e-02) $-$	2.2184e+00 (2.2493e-03) $-$	2.1996e+00 (2.2345e-02) $-$	1.9613e+00 (2.8010e-01) $-$	1.1867e+00 (5.3394e-01)
	50%	3.2896e+00 (1.7396e-01) $-$	3.0538e+00 (2.8879e-02) $-$	3.0190e+00 (5.2262e-02) $-$	2.8556e+00 (2.7824e-01) $-$	2.0908e+00 (7.1504e-01)
	75%	4.5694e+00 (3.5324e-01) $-$	4.0529e+00 (5.4464e-02) $\approx$	4.0215e+00 (1.0754e-01) $\approx$	3.9965e+00 (3.1599e-01) $\approx$	8.3347e+00 (1.0237e+00)
	95%	6.9576e+00 (1.0843e+00) $\approx$	5.5961e+00 (5.3679e-02) $\approx$	5.6836e+00 (2.4876e-01) $\approx$	6.0923e+00 (5.4639e-01) $\approx$	5.6729e+00 (2.1257e+00)
Ackley-I	5%	2.1040e+00 (1.8214e-01) $-$	1.3733e+00 (3.3794e-01) $-$	9.6333e-02 (1.8901e-02) $\approx$	1.7585e-01 (1.2440e-01) $-$	9.7543e-02 (3.9146e-02)
	25%	2.9738e+00 (1.8514e-01) $-$	2.3535e+00 (2.1429e-01) $-$	2.0568e-01 (3.5956e-02) $\approx$	5.2164e-01 (5.8015e-01) $-$	2.0331e-01 (7.4606e-02)
	50%	3.5819e+00 (1.6709e-01) $-$	2.9171e+00 (2.5604e-01) $-$	3.5220e-01 (5.6119e-02) $\approx$	7.7263e-01 (7.7222e-01) $-$	3.2185e-01 (1.1253e-01)
	75%	4.5056e+00 (1.1115e-01) $-$	3.6170e+00 (2.8501e-01) $-$	6.7452e-01 (1.0479e-01) $-$	9.9810e-01 (7.4903e-01) $-$	4.8967e-01 (1.7913e-01)
	95%	6.0673e+00 (2.1242e-01) $-$	5.0808e+00 (1.6938e-01) $-$	2.0580e+00 (1.5660e-02) $-$	1.9350e+00 (1.6146e+00) $-$	9.1072e-01 (4.9060e-01)
Ackley-II	5%	3.6226e+00 (1.7790e-01) $-$	3.7484e+00 (2.4236e-02) $-$	3.7061e+00 (4.6585e-02) $-$	3.4072e+00 (2.8914e-01) $-$	2.5089e+00 (7.6250e-01)
	25%	4.3273e+00 (1.9539e-01) $-$	4.3649e+00 (1.6628e-02) $-$	4.2669e+00 (6.6596e-02) $-$	4.0061e+00 (1.9874e-01) $\approx$	3.5385e+00 (7.6554e-01)
	50%	4.9140e+00 (2.6313e-01) $\approx$	4.8317e+00 (1.2213e-01) $\approx$	4.6222e+00 (6.0964e-02) $\approx$	4.4228e+00 (1.6292e-01) $\approx$	4.3123e+00 (8.1509e-01)
	75%	5.5161e+00 (2.7997e-01) $\approx$	5.4248e+00 (1.3343e-01) $\approx$	5.2040e+00 (1.5259e-01) $\approx$	4.9905e+00 (2.3661e-01) $\approx$	4.9644e+00 (8.1048e-01)
	95%	6.4260e+00 (4.2145e-01) $\approx$	6.0830e+00 (1.8482e-01) $\approx$	5.8336e+00 (1.3509e-01) $\approx$	5.6838e+00 (2.5812e-01) $\approx$	5.8641e+00 (7.5154e-01)
Rastrigin-I	5%	2.6359e+01 (6.2784e+00) $\approx$	3.6730e+01 (7.6639e+00) $-$	2.5169e+01 (2.6266e+00) $\approx$	2.7618e+01 (3.6902e+00) $\approx$	2.6238e+01 (2.5165e+00)
	25%	4.4250e+01 (7.7101e+00) $\approx$	5.7113e+01 (1.2497e+01) $-$	4.2094e+01 (3.1510e+00) $\approx$	4.4928e+01 (5.5935e+00) $\approx$	4.2326e+01 (3.6744e+00)
	50%	5.9753e+01 (1.0151e+01) $-$	7.4654e+01 (1.8622e+01) $-$	5.6112e+01 (3.9190e+00) $\approx$	5.9310e+01 (8.8632e+00) $\approx$	5.4785e+01 (5.4205e+00)
	75%	8.3515e+01 (1.8001e+01) $-$	9.8313e+01 (2.8369e+01) $-$	7.4887e+01 (6.2412e+00) $-$	7.9073e+01 (1.9286e+01) $-$	6.8098e+01 (8.2728e+00)
	95%	1.5818e+02 (4.3552e+01) $-$	1.6669e+02 (4.0435e+01) $-$	1.3682e+02 (2.2979e+01) $-$	1.2808e+02 (5.8301e+01) $-$	8.9259e+01 (1.5389e+01)
Rastrigin-II	5%	6.4737e+01 (1.7582e+00) $-$	6.7883e+01 (1.5605e+00) $-$	6.3057e+01 (2.2005e+00) $-$	6.1093e+01 (2.8075e+00) $-$	4.5054e+01 (6.0348e+00)
	25%	9.7163e+01 (3.5574e+00) $-$	9.4716e+01 (1.7024e+00) $-$	9.3105e+01 (2.6550e+00) $-$	9.6881e+01 (4.1387e+00) $-$	7.1361e+01 (1.1557e+01)
	50%	1.2564e+02 (6.9602e+00) $-$	1.1717e+02 (2.2589e+00) $-$	1.1905e+02 (3.9188e+00) $-$	1.2988e+02 (9.3123e+00) $-$	9.4798e+01 (1.6858e+01)
	75%	1.6150e+02 (1.3363e+01) $-$	1.4235e+02 (2.8370e+00) $-$	1.5109e+02 (6.5386e+00) $-$	1.6934e+02 (1.5657e+01) $-$	1.2334e+02 (2.3236e+01)
	95%	2.2799e+02 (2.9678e+01) $-$	1.8137e+02 (5.4616e+00) $\approx$	2.0929e+02 (1.4155e+01) $\approx$	2.3501e+02 (2.6453e+01) $-$	1.7181e+02 (3.4438e+01)
Griewank-I	5%	1.4295e+00 (3.6791e-01) $\approx$	1.6501e+00 (2.4926e-01) $-$	1.2337e+00 (1.7745e-01) $\approx$	1.5408e+00 (2.4038e-01) $-$	1.3317e+00 (1.5637e-01)
	25%	2.9319e+00 (1.5217e+00) $-$	3.3922e+00 (9.2390e-01) $-$	2.1563e+00 (6.2049e-01) $-$	2.6166e+00 (5.5361e-01) $-$	2.0979e+00 (4.5880e-01)
	50%	5.6758e+00 (3.7501e+00) $-$	5.6437e+00 (1.6059e+00) $-$	3.7437e+00 (1.3372e+00) $\approx$	4.1832e+00 (1.1701e+00) $-$	3.0935e+00 (1.0398e+00)
	75%	1.1592e+01 (7.4868e+00) $-$	1.0284e+01 (2.4744e+00) $-$	7.8351e+00 (2.3734e+00) $-$	7.8516e+00 (3.4542e+00) $-$	4.7885e+00 (1.9504e+00)
	95%	2.8449e+01 (1.2148e+01) $-$	2.7646e+01 (5.1187e+00) $-$	2.4819e+01 (5.0000e+00) $-$	1.9363e+01 (1.1093e+01) $-$	8.5325e+00 (4.3527e+00)
Griewank-II	5%	8.5584e+00 (4.8922e-01) $-$	8.7020e+00 (3.1887e-01) $-$	8.1302e+00 (5.1623e-01) $-$	6.7492e+00 (1.2233e+00) $-$	4.0665e+00 (1.3843e+00)
	25%	1.3381e+01 (6.6195e-01) $-$	1.3495e+01 (1.0838e-01) $-$	1.3655e+01 (1.6315e-01) $-$	1.2651e+01 (1.6883e+00) $-$	9.0974e+00 (3.1722e+00)
	50%	1.8848e+01 (1.3946e+00) $-$	1.8258e+01 (3.7859e-01) $\approx$	1.9214e+01 (5.9426e-01) $-$	1.8282e+01 (2.5765e+00) $\approx$	1.4656e+01 (4.2747e+00)
	75%	2.5659e+01 (2.6930e+00) $\approx$	2.3927e+01 (8.2098e-01) $\approx$	2.6220e+01 (8.5712e-01) $\approx$	2.5084e+01 (3.7183e+00) $-$	2.2094e+01 (5.0576e+00)
	95%	3.7448e+01 (6.2271e+00) $\approx$	3.2592e+01 (1.6257e+00) $\approx$	3.8192e+01 (1.5611e+00) $\approx$	3.7894e+01 (6.7145e+00) $-$	5.5562e+01 (6.5583e+00)

TABLE II
COMPARATIVE RESULTS ON ADAPTIVE CONTROL SYSTEMS OVER 20 INDEPENDENT RUNS

Problems	Quantile	GP-based Optimization	PMTO-FT	(θ_t, θ_u) -PMTO
Robot Arm	5%	4.2821e-02 (1.9260e-02) $-$	3.4766e-02 (7.9264e-03) $-$	1.7336e-02 (5.2244e-03)
	25%	8.9019e-02 (2.5297e-02) $-$	7.6430e-02 (1.7672e-02) $-$	4.7606e-02 (1.4060e-02)
	50%	1.3401e-01 (2.0803e-02) $-$	1.0522e-01 (2.0310e-02) $-$	7.5639e-02 (1.6956e-02)
	75%	1.7645e-01 (1.5028e-02) $-$	1.3607e-01 (1.9514e-02) $\approx$	1.1816e-01 (1.8595e-02)
	95%	2.3556e-01 (1.6606e-02) $-$	1.7845e-01 (1.4280e-02) $\approx$	1.7742e-01 (8.5307e-03)
Crane Load-I	5%	3.0679e+04 (5.7316e+04) $-$	1.4233e+03 (1.3452e+03) $-$	4.6175e+02 (3.2924e+02)
	25%	1.1283e+05 (1.4729e+05) $-$	2.7508e+04 (2.0614e+04) $-$	1.0081e+04 (4.3555e+03)
	50%	2.1993e+05 (2.1653e+05) $-$	1.0485e+05 (7.1414e+04) $-$	3.9601e+04 (1.2591e+04)
	75%	3.7102e+05 (2.9138e+05) $-$	2.6922e+05 (1.6376e+05) $-$	1.4146e+05 (1.2170e+05)
	95%	6.0364e+05 (3.7526e+05) $\approx$	6.1116e+05 (3.8202e+05) $\approx$	3.4578e+05 (3.2520e+05)
Crane Load-II	5%	6.3392e+02 (3.5072e+02) $-$	4.7310e+02 (1.1462e+02) $-$	1.5026e+02 (7.6382e+01)
	25%	1.6577e+04 (9.7445e+03) $-$	1.1846e+04 (3.3891e+03) $-$	3.0218e+03 (9.1541e+02)
	50%	7.4319e+04 (4.7370e+04) $-$	5.7356e+04 (1.8227e+04) $-$	1.1835e+04 (2.1282e+03)
	75%	1.7974e+05 (9.7858e+04) $-$	1.6482e+05 (5.2057e+04) $-$	2.9978e+04 (7.5976e+03)
	95%	3.9408e+05 (2.0022e+05) $-$	3.8901e+05 (1.1719e+05) $-$	8.8104e+04 (3.5100e+04)

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- ❑ Research Topic Introduction
 - ❑ Multiobjective Optimization
 - ❑ Few-Shot Challenges
 - ❑ Multitask Optimization
 - ❑ From Multitask Optimization to Parametric Multitask Optimization
- ❑ Problem 1: Finite Few-Shot Single-Objective Multitask Optimization
 - In-Context Guidance: Learning Inter-Task Synergies via Numerical Foundational Models for Few-Shot Multitask Optimization^[1]
- ❑ Problem 2: Finite Few-Shot Multiobjective Multitask Optimization
 - Bayesian Forward-Inverse Transfer for Multiobjective Optimization^[2]
 - Solving Few-Shot Multiobjective Multitask Optimization via Iterative Sequential Transfer^[3]
- ❑ Problem 3: Parametric Single-Objective Multitask Optimization
 - (θ_l, θ_u) -Parametric Multi-Task Optimization: Joint Search in Solution and Infinite Task Spaces^[4]
- ❑ Problem 4: Parametric Multiobjective Multitask Optimization
 - Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling^[5]
- ❑ Conclusions and Potential Future Work

[1] Tingyang Wei, et al. “In-Context Guidance: Learning Inter-Task Synergies via Numerical Foundational Models for Few-Shot Multitask Optimization.” In submission to IEEE Transactions on Evolutionary Computation (2026)

[2] Tingyang Wei, et al. "Bayesian forward-inverse transfer for multiobjective optimization." International Conference on Parallel Problem Solving from Nature. Cham: Springer Nature Switzerland, 2024.

[3] Tingyang Wei, et al. “Solving Few-Shot Multiobjective Multitask Optimization via Iterative Sequential Transfer.” 2026 IEEE Congress on Evolutionary Computation (CEC).

[4] Tingyang Wei, Jiao Liu, Abhishek Gupta, Puay Siew Tan, Yew-Soon Ong, “ (θ_l, θ_u) -Parametric Multi-Task Optimization: Joint Search in Solution and Infinite Task Spaces” IEEE Transactions on Evolutionary Computation (2025).

[5] Tingyang Wei, Jiao Liu, Abhishek Gupta, Chin Chun Ooi, Puay Siew Tan, Yew-Soon Ong, “Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling”, *IJCAI 2026*.

Parametric Multiobjective Multitask Optimization

$$\begin{aligned} \operatorname{argmin}_{x \in \mathcal{X}} F(x) = \\ \operatorname{argmin}_{x \in \mathcal{X}} (f_1(x), \dots, f_M(x)) \end{aligned}$$

- Multitask Optimization (MOO):

- $\operatorname{argmin}_{x_m \in \mathcal{X}_m} F_m(x_m), m \in [M]$
- Finite MOO Tasks

Assuming K multi-objective problems share certain similarities, we use their **Pareto-optimal solution distributions and surrogate models** as transferable knowledge to accelerate convergence.

- Parametric Multitask Optimization:

- $\operatorname{argmin}_{x \in \mathcal{X}} F(x, \theta), \forall \theta \in \Theta$
- Infinite MOO Tasks

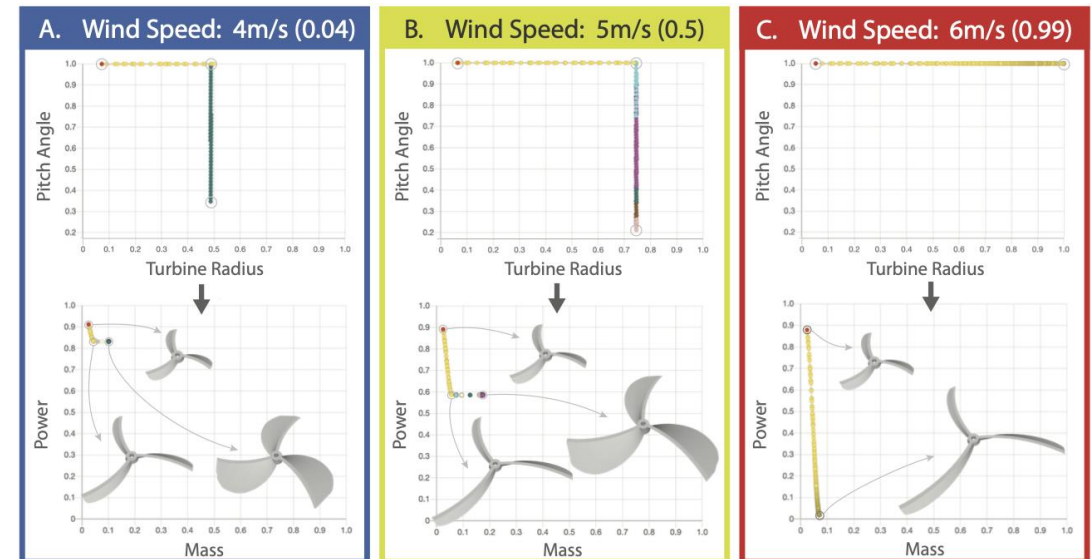
We assume access to **an external task space Θ** and investigate how leveraging **task parameters θ as side information** can accelerate optimization convergence and **return any preferred solution observing preference vector λ in any task θ .**

Why Parametric Multiobjective Multitask Optimization?

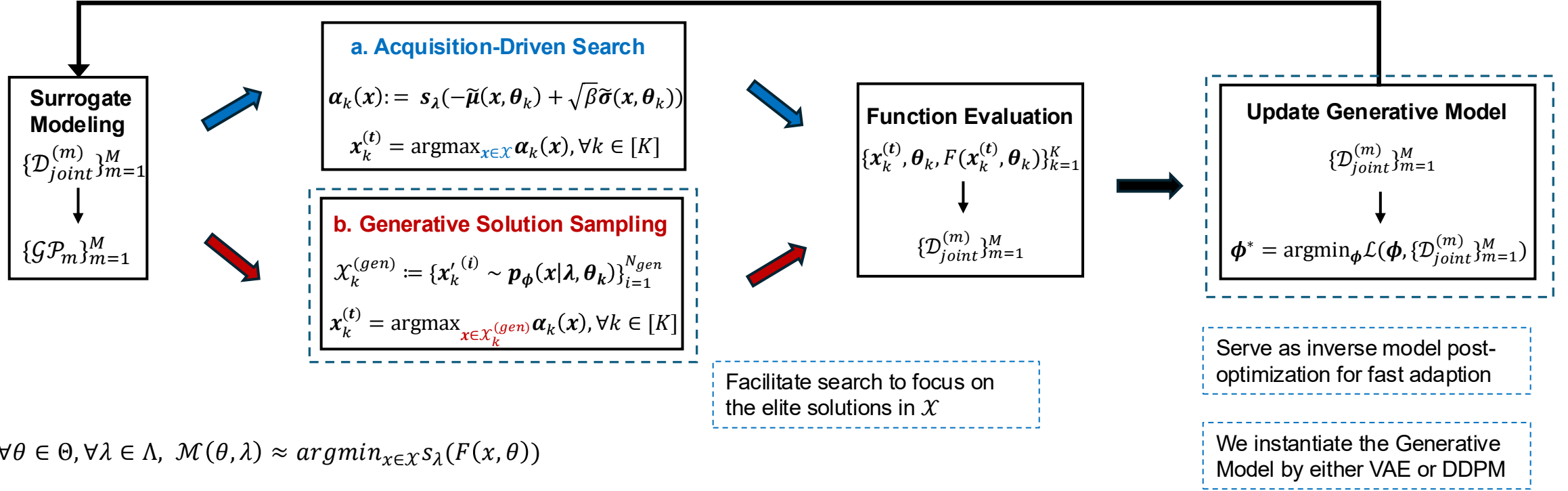
- $\operatorname{argmin}_{x \in \mathcal{X}} F(x, \theta) := (f_1(x, \theta), \dots, f_M(x, \theta)), \forall \theta \in \Theta$
- Opportunities:
 - Explicit task parameters θ provide a structured way to **quantify task similarity**, contributing to **faster multi-objective optimization convergence** via efficient knowledge transfer.
- Challenges:
 - The task parameters θ lie in a continuous space, meaning we have **infinitely many** multi-objective tasks to solve, and each multi-objective task require **potentially infinitely many solutions** to satisfy **arbitrary preference articulation** on Pareto Front for **any task**.
- Application Example:
 - In turbine design, θ can represent different **wind speeds**. Optimizing manufactured components x for diverse θ allows the parametric turbine to **adapt to unexpected environments**.

$$\forall \theta \in \Theta, \forall \lambda \in \Lambda$$
$$\mathcal{M}(\theta, \lambda) \approx \operatorname{argmin}_{x \in \mathcal{X}} s_{\lambda}(F(x, \theta))$$

λ means preference vector to trade-off the objectives



Methodology



- $\forall \theta \in \Theta, \forall \lambda \in \Lambda, \mathcal{M}(\theta, \lambda) \approx \operatorname{argmin}_{x \in \mathcal{X}} s_\lambda(F(x, \theta))$

- Surrogate Models:

- $\mathcal{GP}: \mathcal{X} \times \Theta \rightarrow \mathbb{R}$
- (M models for M objectives)

Facilitate cross-task exploration in $\mathcal{X} \times \Theta$

- Cross-Task Sampling:

- Forward Model contains cross-task information:
- $\mathcal{GP}(\mu(\cdot, \theta), \sigma^2(\cdot, \theta))$

- Conditional Generative Model (Inverse Model):

- $\mathcal{M}_\phi: \Theta \times \Lambda \rightarrow \mathcal{X}$
- (Identify the preferred solution by λ for task θ)

- Alternating Optimization Loop:

- Per iteration, mode **a.** and **b.** alternates in a round robin way

Results

- $\operatorname{argmin}_{x \in \mathcal{X}} F(x, \theta) := (f_1(x, \theta), \dots, f_M(x, \theta)), \forall \theta \in \Theta$
- Good Optimization Results
 - Sample K multi-objective tasks
 - Check out the multi-objective convergence
 - $F(\cdot, \theta_k), k \in [K]$
 - Theoretical Analysis of faster convergence

Theorem 2 (Tightened Regret Bounds for PMT-MOBO). Under Assumption 1 and with the task-aware kernel that satisfies: $\tilde{\kappa}_m((\mathbf{x}, \theta_k), (\mathbf{x}', \theta_k)) = \kappa_m(\mathbf{x}, \mathbf{x}'), \forall k \in [K]$, let $\tilde{\gamma}_{T,k} = \max_m \tilde{\gamma}_{T,m,k}$ be the largest MIG of PMT-MOBO across M objective functions over T scalarized evaluations for the k -th task, it holds that

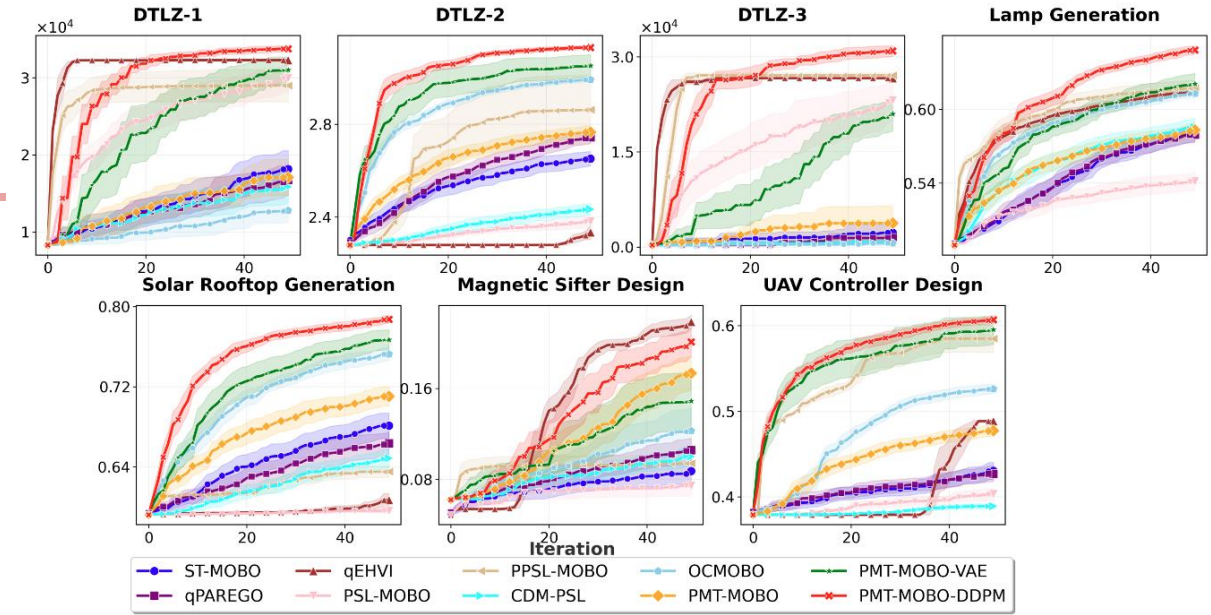
$$\tilde{\gamma}_{T,m,k} \leq \gamma_{T,m,k}, \forall m \in [M], \forall k \in [K]. \quad (13)$$

Accordingly, it also holds that $\tilde{\gamma}_{T,k} \leq \gamma_{T,k}, \forall k \in [K]$. Consider that both regret bounds from Theorem 1 still hold with $\gamma_{T,k}$ replaced by $\tilde{\gamma}_{T,k}$, consequently, it holds that,

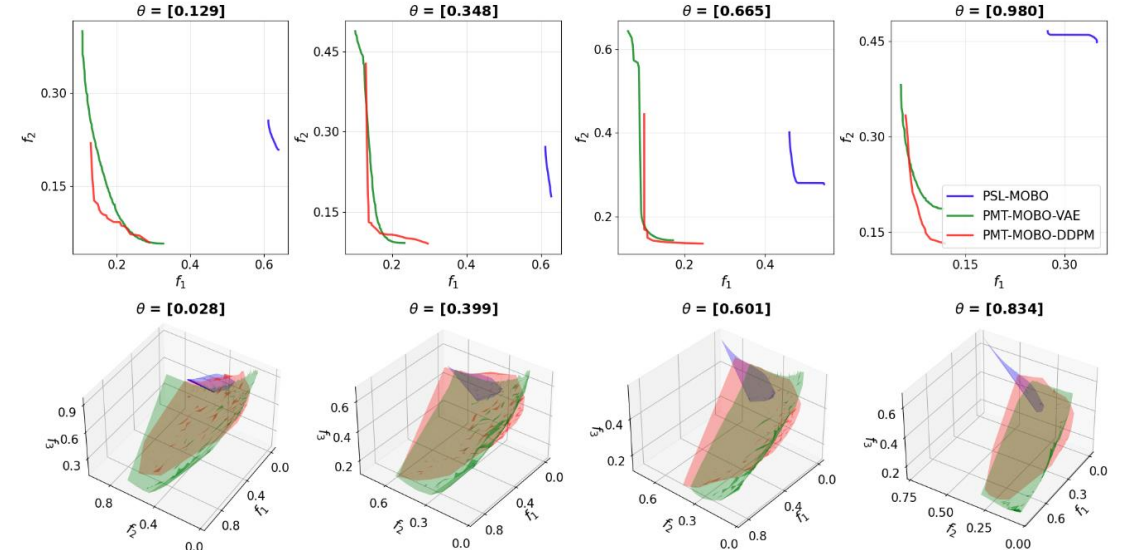
$$\mathbb{E}[\mathcal{R}_B^{\text{PMT}}(T)] \leq \mathbb{E}[\mathcal{R}_B(T)], \quad (14)$$

i.e. PMT-MOBO achieves a tighter upper bound on Bayes regret than standard ST-MOBO.

- $\forall \theta \in \Theta, \forall \lambda \in \Lambda, \mathcal{M}(\theta, \lambda) \approx \operatorname{argmin}_{x \in \mathcal{X}} s_\lambda(F(x, \theta))$
- Good Inverse Model Predictive Results
 - Sample multiple (unseen) multi-objective tasks θ
 - Sample diverse preference vectors $\lambda \in \Lambda$
 - Check out the solution quality of the model output
 - $F(x', \theta), x' = \mathcal{M}_\phi(\theta, \lambda),$



Our PMT-MOBO-VAE/DDPM exhibit better hypervolume results



Our PMT-MOBO-VAE/DDPM generate better Pareto-optimal solutions

Detailed Results

Method	DTLZ-1	DTLZ-2	DTLZ-3	LAMP	SOLAR	UAV
PSL-MOBO	2.82e+04 (1.78e+03) [≈] ---	1.90e+00 (1.30e-01) ⁻⁻⁻	2.39e+04 (3.22e+03) ⁺⁻⁻	5.01e-01 (7.32e-02) ⁻⁻⁻	3.54e-01 (4.06e-02) ⁻⁻⁻	4.12e-01 (7.30e-03) ⁻⁻⁻
CDM-PSL	1.92e+04 (3.85e+03) ⁻⁻⁻	2.52e+00 (8.49e-02) ⁻⁻⁻	1.83e+03 (1.31e+03) ⁻⁻⁻	6.19e-01 (4.45e-02) [≈] ---	6.54e-01 (3.11e-02) ⁻⁻⁻	4.26e-01 (3.09e-02) ⁻⁻⁻
PPSL-MOBO	2.24e+04 (1.10e+02) ⁻⁻⁻	2.98e+00 (3.56e-02) ⁻⁻⁻	2.67e+04 (6.08e+02) ⁺⁻⁻	6.12e-01 (1.05e-01) [≈] ---	4.86e-01 (1.11e-01) ⁻⁻⁻	5.24e-01 (1.71e-01) ⁻⁻⁻
PMT-MOBO-VAE	2.88e+04 (2.57e+03)	3.10e+00 (2.99e-02)	1.78e+04 (7.28e+03)	6.31e-01 (7.33e-02)	7.93e-01 (1.56e-02)	6.08e-01 (1.88e-02)
PMT-MOBO-DDPM	3.47e+04 (4.85e+02)	3.13e+00 (1.41e-02)	3.20e+04 (1.98e+03)	6.70e-01 (5.82e-02)	7.65e-01 (1.30e-02)	6.05e-01 (2.44e-02)

Table 1: Inverse Model Generalization Performance on Unseen EMOP Tasks. Statistical significance is assessed using the Wilcoxon signed-rank test at the 95% confidence level. Superscripts ^{s1s2} indicate that the competitor is significantly worse (−), significantly better (+), or statistically similar (≈) relative to **PMT-MOBO-VAE** (*s1*) and **PMT-MOBO-DDPM** (*s2*).

Method	DTLZ-1	DTLZ-2	DTLZ-3	LAMP	SOLAR	MAGNETIC	UAV
ST-MOBO	1.82e+04 (4.80e+03) ⁻⁻⁻	2.65e+00 (6.30e-02) ⁻⁻⁻	2.24e+03 (3.03e+03) ⁻⁻⁻	5.80e-01 (1.20e-02) ⁻⁻⁻	6.81e-01 (2.51e-02) ⁻⁻⁻	8.72e-02 (2.15e-02) ⁻⁻⁻	4.31e-01 (1.99e-02) ⁻⁻⁻
qPAREGO	1.68e+04 (4.03e+03) ⁻⁻⁻	2.74e+00 (5.30e-02) ⁻⁻⁻	1.42e+03 (1.48e+03) ⁻⁻⁻	5.79e-01 (1.30e-02) ⁻⁻⁻	6.64e-01 (2.21e-02) ⁻⁻⁻	1.06e-01 (2.05e-02) ⁻⁻⁻	4.27e-01 (1.82e-02) ⁻⁻⁻
qEHVI	3.23e+04 (5.06e+01) [≈] ---	2.33e+00 (6.20e-02) ⁻⁻⁻	2.66e+04 (1.85e-02) ⁺	6.14e-01 (3.00e-03) ⁻⁻⁻	6.07e-01 (1.29e-02) ⁻⁻⁻	2.18e-01 (1.15e-02) ⁺⁺	4.89e-01 (2.60e-03) ⁻⁻⁻
PSL-MOBO	3.00e+04 (3.52e+03) [≈] ---	2.38e+00 (4.90e-02) ⁻⁻⁻	2.31e+04 (6.44e+03) [≈] ---	5.42e-01 (1.40e-02) ⁻⁻⁻	5.96e-01 (3.60e-03) ⁻⁻⁻	7.42e-02 (1.79e-02) ⁻⁻⁻	4.03e-01 (1.46e-02) ⁻⁻⁻
CDM-PSL	1.59e+04 (4.15e+03) ⁻⁻⁻	2.43e+00 (5.04e-02) ⁻⁻⁻	7.76e+02 (9.41e+02) ⁻⁻⁻	5.85e-01 (1.39e-02) ⁻⁻⁻	6.49e-01 (1.97e-02) ⁻⁻⁻	1.00e-01 (1.82e-02) ⁻⁻⁻	3.89e-01 (5.70e-03) ⁻⁻⁻
PPSL-MOBO	3.20e+04 (1.72e+03) [≈] ---	2.86e+00 (3.00e-01) [≈] ---	2.71e+04 (7.89e+02) ⁺	6.09e-01 (1.61e-02) ⁻⁻⁻	6.35e-01 (1.15e-02) ⁻⁻⁻	9.41e-02 (1.69e-02) ⁻⁻⁻	5.85e-01 (3.19e-02) [≈] ---
OCMOBO	1.28e+04 (3.04e+03) ⁻⁻⁻	2.99e+00 (2.12e-02) [≈] ---	6.20e+02 (6.29e+02) ⁻⁻⁻	6.13e-01 (6.60e-03) ⁻⁻⁻	7.53e-01 (1.36e-02) ⁻⁻⁻	1.23e-01 (3.95e-02) ⁻⁻⁻	5.26e-01 (1.09e-02) ⁻⁻⁻
PMT-MOBO	1.71e+04 (4.70e+03) ⁻⁻⁻	2.77e+00 (5.50e-02) ⁻⁻⁻	3.78e+03 (5.40e+03) ⁻⁻⁻	5.83e-01 (1.22e-02) ⁻⁻⁻	7.10e-01 (1.90e-02) ⁻⁻⁻	1.73e-01 (2.99e-02) [≈] ---	4.78e-01 (1.27e-02) ⁻⁻⁻
PMT-MOBO-VAE	3.10e+04 (3.41e+03)	3.05e+00 (9.58e-02)	2.10e+04 (5.33e+03)	6.21e-01 (1.73e-02)	7.67e-01 (2.01e-02)	1.49e-01 (5.07e-02)	5.95e-01 (3.33e-02)
PMT-MOBO-DDPM	3.37e+04 (8.35e+02)	3.13e+00 (1.62e-02)	3.09e+04 (1.67e+03)	6.48e-01 (5.10e-03)	7.87e-01 (7.80e-03)	2.01e-01 (2.50e-02)	6.07e-01 (7.50e-03)

Table 5: Average Hypervolume results (mean $\pm$ std) across synthetic and real-world benchmarks. Statistical significance is assessed using the Wilcoxon signed-rank test at the 95% confidence level. Superscripts ^{s1s2} indicate that the competitor is significantly worse (−), significantly better (+), or statistically similar (≈) relative to **PMT-MOBO-VAE** (*s1*) and **PMT-MOBO-DDPM** (*s2*).

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 - Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling^[5]
- ❑ Conclusions and Potential Future Work

[1] Tingyang Wei, et al. “In-Context Guidance: Learning Inter-Task Synergies via Numerical Foundational Models for Few-Shot Multitask Optimization.” In submission to IEEE Transactions on Evolutionary Computation (2026)

[2] Tingyang Wei, et al. "Bayesian forward-inverse transfer for multiobjective optimization." International Conference on Parallel Problem Solving from Nature. Cham: Springer Nature Switzerland, 2024.

[3] Tingyang Wei, et al. “Solving Few-Shot Multiobjective Multitask Optimization via Iterative Sequential Transfer.” 2026 IEEE Congress on Evolutionary Computation (CEC).

[4] Tingyang Wei, Jiao Liu, Abhishek Gupta, Puay Siew Tan, Yew-Soon Ong, “ (θ_l, θ_u) -Parametric Multi-Task Optimization: Joint Search in Solution and Infinite Task Spaces” IEEE Transactions on Evolutionary Computation (2025).

[5] Tingyang Wei, Jiao Liu, Abhishek Gupta, Chin Chun Ooi, Puay Siew Tan, Yew-Soon Ong, “Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling”, *IJCAI 2026*.

Retrospect the Thesis Structure

Topic	Relevant Paper
Finite Few-Shot Single-Objective Multitask Optimization	[1]
Finite Few-Shot Multiobjective Multitask Optimization	[2,3]
Parametric Single-Objective Multitask Optimization	[4]
Parametric Multiobjective Multitask Optimization	[5]

[1] Tingyang Wei, et al. “In-Context Guidance: Learning Inter-Task Synergies via Numerical Foundational Models for Few-Shot Multitask Optimization.” In submission to IEEE Transactions on Evolutionary Computation (2026).

[2] Tingyang Wei, Jiao Liu, Abhishek Gupta, Puay Siew Tan, Yew-Soon Ong, “Bayesian Forward-Inverse Transfer for Multiobjective Optimization” International Conference on Parallel Problem Solving from Nature 2024.

[3] Tingyang Wei, Haofeng Wu, Ananda Phan Iman, Zhao Wei, Jiao Liu, Yew-Soon Ong, “Solving Few-Shot Multiobjective Multitask Optimization via Iterative Sequential Transfer.” 2026 IEEE Congress on Evolutionary Computation (CEC).

[4] Tingyang Wei, Jiao Liu, Abhishek Gupta, Puay Siew Tan, Yew-Soon Ong, “ (θ_l, θ_u) -Parametric Multi-Task Optimization: Joint Search in Solution and Infinite Task Spaces” IEEE Transactions on Evolutionary Computation (2025).

[5] Tingyang Wei, Jiao Liu, Abhishek Gupta, Chin Chun Ooi, Puay Siew Tan, Yew-Soon Ong, “Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling”, *IJCAI 2026*.

Limitations

- GP-based surrogate modeling may become expensive in large-data settings.

- We adopt GP as the backbone for its rich theoretical analysis background.
- The bottleneck of GP is the training complexity of $O(N^3)$, where N is the training sample size.

- It is unclear how to theoretically justify the effectiveness of generative solution modeling in the MOO methods.

- Generative solution modeling makes sense in MOO problem due to some common assumptions as below:
 - Under mild conditions, a Pareto set is an $(M-1)$ -dimensional manifold, given that M is the number of the objectives^[1].
- However, it is unclear why involving generative models can further enhance the performance of multiobjective multitask optimization from the perspective of convergence.
 - We only conduct the convergence analysis for method without generative models.

Theorem. Let the kernel functions used in PMTO-FT and the independent strategy satisfy $\kappa_{pmt}((\mathbf{x}, \boldsymbol{\theta}), (\mathbf{x}', \boldsymbol{\theta})) = \kappa_{ind}(\mathbf{x}, \mathbf{x}')$. Then, the MIG in the independent strategy, denoted as $\gamma_{T,m}^{ind}$, and the MIG in the PMTO-FT, denoted as $\gamma_{T,m}$, satisfy $\gamma_{T,m} \leq \gamma_{T,m}^{ind}, \forall m \in [M], \forall T \geq 1$.

[1] Cheng, Ran, et al. "A multiobjective evolutionary algorithm using Gaussian process-based inverse modeling." IEEE Transactions on Evolutionary Computation 19.6 (2015): 838-856..

Limitations -> Future Directions

- GP-based surrogate modeling may become expensive in large-data settings.

- We adopt GP as the backbone for its rich theoretical analysis background.
- The bottleneck of GP is the training complexity of $O(N^3)$, where N is the training sample size.

(A more scalable surrogate model supporting large-data settings and theoretical analysis).

- It is unclear how to theoretically justify the effectiveness of generative solution modeling in the MOO methods.

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 - Under mild conditions, a Pareto set is an $(M-1)$ -dimensional manifold, given that M is the number of the objectives^[1].
- However, it is unclear why involving generative models can further enhance the performance of multiobjective multitask optimization from the perspective of convergence.
 - We only conduct the convergence analysis for method without generative models.

(A more direct convergence analysis for MOO methods involving generative modeling).

Theorem. Let the kernel functions used in PMTO-FT and the independent strategy satisfy $\kappa_{pmt}((\mathbf{x}, \boldsymbol{\theta}), (\mathbf{x}', \boldsymbol{\theta})) = \kappa_{ind}(\mathbf{x}, \mathbf{x}')$. Then, the MIG in the independent strategy, denoted as $\gamma_{T,m}^{ind}$, and the MIG in the PMTO-FT, denoted as $\gamma_{T,m}$, satisfy $\gamma_{T,m} \leq \gamma_{T,m}^{ind}, \forall m \in [M], \forall T \geq 1$.



Thank you!